ISSN 0040-5795, Theoretical Foundations of Chemical Engineering, 2009, Vol. 43, No. 5, pp. 642–662. © Pleiades Publishing, Ltd., 2009.
Original Russian Text © S.N. Aristov, D.V. Knyazev, A.D. Polyanin, 2009, published in Teoreticheskie Osnovy Khimicheskoi Tekhnologii, 2009, Vol. 43, No. 5, pp. 547–566.

Exact Solutions of the Navier–Stokes Equations with the Linear Dependence of Velocity Components on Two Space Variables

S. N. Aristov[a], D. V. Knyazev[a], and A. D. Polyanin[b]

[a] *Institute of Continuous Media Mechanics, Ural Branch, Russian Academy of Sciences, Perm, Russia*

[b] *Institute for Problems in Mechanics, Russian Academy of Sciences, Moscow, Russia*

e-mail: polyanin@ipmnet.ru

Received April 9, 2009

Abstract—A wide class of two-dimensional and three-dimensional steady-state and non-steady-state flows of a viscous incompressible fluid is considered. It is assumed that the components of the velocity of a fluid linearly depend on two spatial coordinates. The three-dimensional Navier–Stokes equations in this case are reduced to a closed determining system that consists of six equations with partial derivatives of the third and second orders. A brief review of the known exact solutions of this system and the respective flows of a fluid (Couette–Poiseuille, Ekman, Stokes, Karman, and other flows) is given. The cases of reducing a determining system to one or two equations are described. Many new exact solutions of two-dimensional and three-dimensional nonstationary Navier–Stokes equations containing arbitrary functions and arbitrary parameters are derived. Periodic (both in spatial coordinates and in time) and some other solutions that are expressed in terms of elementary functions are described. The problems of the nonlinear stability of solutions are studied. A number of new hydrodynamic problems are considered. A general interpretation of the solutions as the main terms of the Taylor series expansion in terms of radial coordinates is given.

DOI: 10.1134/S0040579509050066

INTRODUCTION

Exact solutions of the Navier–Stokes equations favor a better understanding of the qualitative features of steady-state and non-steady-state flows (stability, nonuniqueness, blow-up regimes, etc.). They make it possible to estimate the applicability area of simplified hydrodynamic models (a nonviscous fluid, creeping flows, boundary layer, etc.) and are indispensable for testing the relevant numerical, asymptotic, and approximate analytical methods (it should be noted that even the solutions that do not allow for a clear physical interpretation can be used for testing). The exact solutions of hydrodynamic equations are used for the mathematical modeling of many processes of chemical and petrochemical technologies [1, 2], including the processes of convective heat and mass transfer, and various natural phenomena. The exact solutions of the Navier–Stokes equations are of great applied importance: as vivid examples, it is sufficient to recall the Poiseuille formula, which is widely used to describe flows in tubes, a Couette viscometer, the Karman solution (which is the basis of the theory of a rotating disk Levich electrode), etc. Simple solutions are frequently used to illustrate the theoretical material and some applications in the educational courses of universities and technical institutes of higher education (on the theory of convective heat and mass transfer, chemical engineering, hydro- and aerodynamics, etc. [3–6]).

The search for the exact solutions of the Navier–Stokes equations is a very difficult task due to their nonlinearity. It is convenient to divide all exact solutions of the Navier–Stokes equations (the fullest, but far from exhausting, list of exact solutions is given in [7–9]) into three categories by the appropriate choice of Cartesian coordinates (the simplest classification based on the functional structure of the components of the velocity of a fluid): 1) solutions that are linear with respect to space variables (the greatest number of such solutions is known), 2) solutions that describe conical flows in which velocity decreases in inverse proportion to the distance from the origin of coordinates, and 3) other solutions.

Remark. Plane flows (all quantities depend on x, y, and t) in such a classification fall within the first category, since velocity components can be represented as functions that are linear with respect to z:

$$V_i = V_{i0}(x,y,t) + V_{i1}(x,y,t)z, \quad i = 1,2, V_3 = 0,$$

at $V_{i1}(x, y, t) \equiv 0$ (degenerate case).

In this paper, a wide class of the known and new exact solutions of the stationary and nonstationary (two-dimensional and three-dimensional) Navier–

Stokes equations with the linear dependence of velocity components on two space variables is described from unified positions. In the case under consideration, the Navier–Stokes equations are reduced to a closed determining system that consists of six equations with partial derivatives of the third and second orders. These equations include only two independent variables; therefore, they are simpler than the original Navier–Stokes equations (which include four independent variables). This class of solutions encompasses a considerable portion of solutions from the first category (here, the classification described above is used).

Remark. In published data, there are also other classifications of the exact solutions of the Navier–Stokes equations [7–11]. For example, the two simplest classifications based on the number of independent variables are as follows: 1) stationary and nonstationary solutions [9] and 2) two-dimensional (plane, axisymmetric, etc.) and three-dimensional solutions [10, 11]. The specified classifications can easily be united. For the group classification of the solutions of the Navier–Stokes equations, see [7, 12].

By the exact solutions of the Navier–Stokes equations, the following solutions are meant: 1) solutions that are expressed through elementary functions, 2) solutions that are expressed as quadratures, 3) solutions that are described by ordinary differential equations (systems of ordinary differential equations), and 4) solutions that are expressed through the solutions of linear equations with partial derivatives (linear integral equations).

Further, the main attention is paid to solutions from items 1 and 2.

Solutions that simultaneously satisfy the Euler equations for an ideal fluid (i.e., Eq. (1) at $\nu = 0$) are called the degenerate (nonviscous) solutions of the Navier–Stokes equations. Degenerate solutions are not considered, as a rule, further.

GENERAL PROPERTIES OF THE NAVIER–STOKES EQUATIONS

The three-dimensional non-steady-state flows of a viscous incompressible fluid are described by Navier–Stokes and continuity equations:

$$\frac{\partial V_i}{\partial t} + V_1\frac{\partial V_i}{\partial x} + V_2\frac{\partial V_i}{\partial y} + V_3\frac{\partial V_i}{\partial z} = -\nabla_i P + \nu\left(\frac{\partial^2 V_i}{\partial x^2} + \frac{\partial^2 V_i}{\partial y^2} + \frac{\partial^2 V_i}{\partial z^2}\right), \quad (1)$$

$$\frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z} = 0, \quad i = 1,2,3.$$

Here, x, y, and z are the Cartesian coordinates; t is time; V_1, V_2, and V_3 are the components of the velocity of a fluid; P is the pressure that is referred to the constant density and includes the potential of mass forces; ν is the kinematic viscosity; $\nabla_1 P = \partial P/\partial x$, $\nabla_2 P = \partial P/\partial y$, $\nabla P = \partial P/\partial z$.

The general properties of the solutions of the Navier–Stokes equations that are the corollary of the relevant group analysis [7, 12] and make it possible to "reproduce" exact solutions are described below.

Let $\bar{V}_1(x, y, z, t)$, $\bar{V}_2(x, y, z, t)$, $\bar{V}_3(x, y, z, t)$, and $\bar{P}(x, y, z, t)$ be some solution of the Navier–Stokes equations (1). Then, the following statements are valid:

1. The set of functions

$$V_i = A\bar{V}_i(Ax + B_1, Ay + B_2, Az + B_3, A^2 t + B_4), \quad i = 1,2,3,$$
$$P = A^2\bar{P}(Ax + B_1, Ay + B_2, Az + B_3, A^2 t + B_4) + C,$$

where A, B_1, B_2, B_3, B_4, and C are arbitrary constants, also yields the solution of Eq. (1) (this property is the corollary of the invariance of the equations under a shift by arbitrary constants with respect to all independent variables and the invariance of the equations under coordinated stretching with respect to independent and dependent variables).

2. The set of functions

$$V_1 = \bar{V}_1(X,Y,z,t)\cos\alpha - \bar{V}_2(X,Y,z,t)\sin\alpha,$$
$$V_2 = \bar{V}_1(X,Y,z,t)\sin\alpha + \bar{V}_2(X,Y,z,t)\cos\alpha,$$
$$V_3 = \bar{V}_3(X,Y,z,t), \quad P = \bar{P}(X,Y,z,t),$$
$$X = x\cos\alpha + y\sin\alpha, \quad Y = y\cos\alpha - x\sin\alpha,$$

where α is an arbitrary constant, also yields the solution of Eq. (1) (this property is the corollary of the invariance of the equations under the rotation of the system of coordinates about the z axis by the arbitrary angle α in the (x, y) plane under conditions of the coordinated rotation of velocity components; similar formulas are derived by means of rotations in other planes).

3. The set of functions

$$V_1 = \bar{V}_1(x - x_0, y - y_0, z - z_0, t) + x_0',$$
$$V_2 = \bar{V}_2(x - x_0, y - y_0, z - z_0, t) + y_0',$$
$$V_3 = \bar{V}_3(x - x_0, y - y_0, z - z_0, t) + z_0', \quad (2)$$
$$P = \bar{P}(x - x_0, y - y_0, z - z_0, t) - x_0'' x - y_0'' y - z_0'' z + p_0,$$

where $x = x_0(t)$, $y = y_0(t)$, $z = z_0(t)$, and $p = p_0(t)$ are the arbitrary time functions (primes denote derivatives with respect to t), also yields the solution of Eq. (1) (this property is the corollary of the invariance of the equations under the transition to the system of coordinates, the origin of which moves according to the law $x = x_0(t)$, $y = y_0(t)$, $z = z_0(t)$ and the axes move in parallel to the original axes; in this case, pressure varies in coordination).

4. For plane two-dimensional flows (the quantities V_1, V_2, and P do not depend on z, $V_3 \equiv 0$), the components of velocity can be expressed through a stream function: $V_1 = \partial\Psi/\partial y$ and $V_2 = -\partial\Psi/\partial x$. In this case, the set of functions

$$V_1 = \bar{V}_1(X,Y,t)\cos(\omega t) - \bar{V}_2(X,Y,t)\sin(\omega t) - \omega Y,$$
$$V_2 = \bar{V}_1(X,Y,t)\sin(\omega t) + \bar{V}_2(X,Y,t)\cos(\omega t) - \omega X,$$
$$P = \bar{P}(X,Y,t) - 2\omega\bar{\Psi}(X,Y,t) + \frac{\omega^2}{2}(x^2+y^2),$$
$$X = x\cos(\omega t) + y\sin(\omega t),$$
$$Y = y\cos(\omega t) - x\sin(\omega t),$$

where ω is an arbitrary constant, also yields the solution of Eq. (1) (this property is the corollary of the invariance of the equations with respect to the system of coordinates that rotates about the z axis at a constant angular velocity ω under conditions of the coordinated rotation of the components of velocities and the appropriate variation of pressure).

Properties 3 and 4 make it possible to obtain some nonstationary solutions using stationary solutions.

EXACT SOLUTIONS WITH A LINEAR DEPENDENCE OF VELOCITY COMPONENTS ON TWO SPACE VARIABLES

Structure of solutions. Determining system of equations. System of equations (1) admits a wide class of exact solutions with the linear dependence of the components of velocity on two space variables of the following form:

$$V_i = f_i(z,t)x + g_i(z,t)y + h_i(z,t), \quad i = 1,2; \qquad V_3 = F(z,t). \tag{3}$$

The original idea to search for the exact solutions of the Navier–Stokes equations in form (3), apparently, belongs to Lin [13] (also see [14]). As a result of the substitution of expressions (3) into Eqs. (1) with the subsequent elimination of pressure, polynomial expressions of the first degree arise with respect to the coordinates x and y, such as $A_n x + B_n y + C_n = 0$, where the coefficients A_n, B_n, and C_n depend only on the variables z and t and are expressed through the functions f_i, g_i, h_i, and F and their derivatives. Equating A_n, B_n, and C_n to zero, we obtain an overdetermined system of equations for f_i, g_i, h_i, and F. The analysis of this system ultimately yields the following representation for the components of velocity and pressure:

$$\begin{aligned} V_1 &= U + \left(w - \frac{1}{2}\frac{\partial F}{\partial z}\right)x + vy, \\ V_2 &= V + ux - \left(w + \frac{1}{2}\frac{\partial F}{\partial z}\right)y, \quad V_3 = F, \\ P &= p_0 - \frac{\alpha}{2}x^2 - \frac{\beta}{2}y^2 - \gamma xy - \varepsilon x - \delta y - \frac{1}{2}F^2 \\ &\quad - \int\frac{\partial F}{\partial t}dz + \nu\frac{\partial F}{\partial z}, \end{aligned} \tag{4}$$

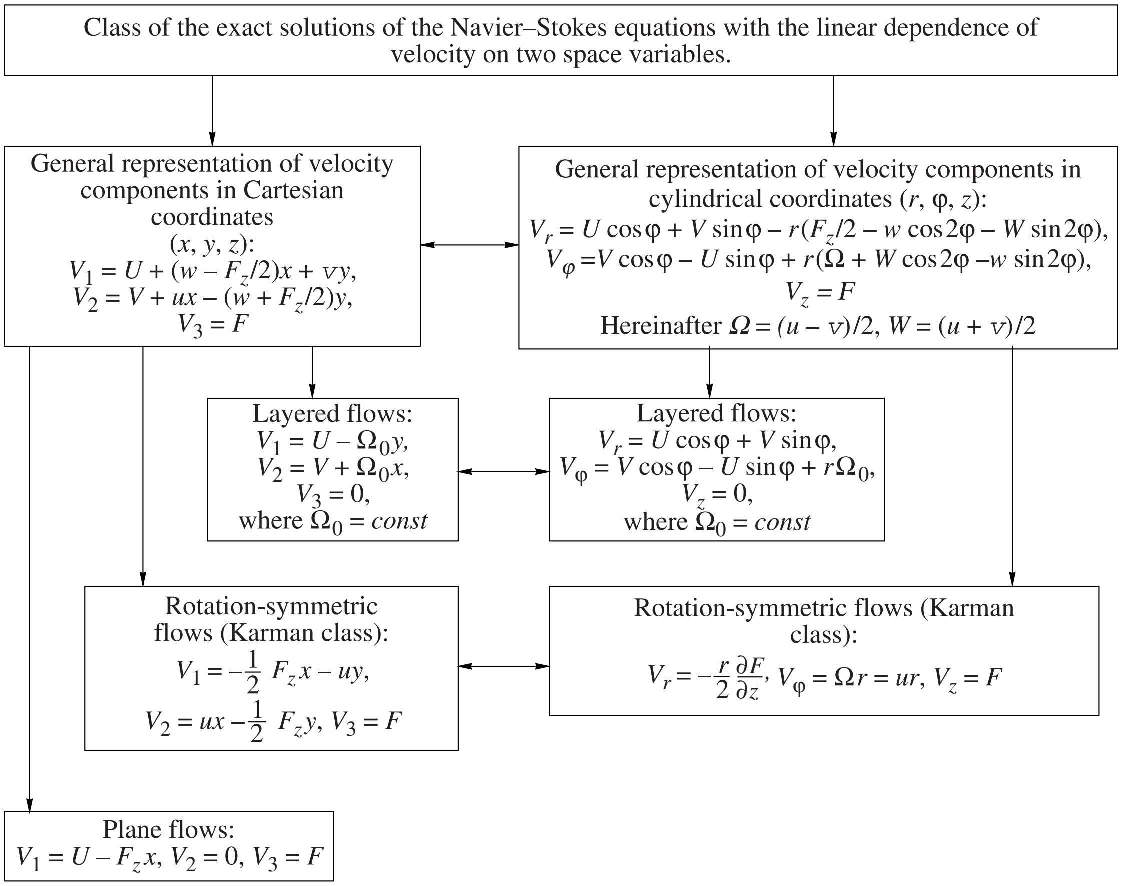
where p_0, α, β, γ, ε, and δ are the arbitrary functions of the time t that determine the radial pressure distribution and U, V, F, u, v, and w are the unknown functions that depend on the coordinate z and the time t.

The expressions for the components of velocity and pressure (4) reduce hydrodynamic equations (1) to a system of six equations with respect to six unknowns U, V, F, u, v, and w that contains five arbitrary time functions α, β, γ, ε, and δ as parameters:

$$\begin{aligned} &\frac{\partial^2 F}{\partial t\partial z} + F\frac{\partial^2 F}{\partial z^2} - \frac{1}{2}\left(\frac{\partial F}{\partial z}\right)^2 \\ &= \nu\frac{\partial^3 F}{\partial z^3} + 2(uv + w^2) - (\alpha+\beta), \\ &\frac{\partial u}{\partial t} + F\frac{\partial u}{\partial z} - u\frac{\partial F}{\partial z} = \nu\frac{\partial^2 u}{\partial z^2} + \gamma, \\ &\frac{\partial v}{\partial t} + F\frac{\partial v}{\partial z} - v\frac{\partial F}{\partial z} = \nu\frac{\partial^2 v}{\partial z^2} + \gamma, \\ &\frac{\partial w}{\partial t} + F\frac{\partial w}{\partial z} - w\frac{\partial F}{\partial z} = \nu\frac{\partial^2 w}{\partial z^2} + \frac{\alpha-\beta}{2}, \\ &\frac{\partial U}{\partial t} + F\frac{\partial U}{\partial z} - U\left(\frac{1}{2}\frac{\partial F}{\partial z} - w\right) + vV = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon, \\ &\frac{\partial V}{\partial t} + F\frac{\partial V}{\partial z} - V\left(\frac{1}{2}\frac{\partial F}{\partial z} + w\right) + uU = \nu\frac{\partial^2 V}{\partial z^2} + \delta. \end{aligned} \tag{5}$$

It should be noted that at $\varepsilon = \delta = \gamma = 0$ (α and β are any), the structure of exact solution (4) and system (5) were derived using other considerations (by studying one class of partially invariant solutions) in [15] and [16], respectively.

Determining system of equations (5) is the subject of the further analysis. Note that the first four equations in (5) are a closed subsystem that is solved irrespective of two last equations.



Scheme of the classification of flows described by system (5) that is based on the structure of the components of the velocity of a fluid. In the general case, the unknown su, v, w, U, V, and F depend on the coordinate z and time t.

Transformation maintaining the form of system (5). If $F_0(z, t)$, $u_0(z, t)$, $v_0(z, t)$, $w_0(z, t)$, $U_0(z, t)$, $V_0(z, t)$ is some solution of system (5), a set of functions

$$\begin{aligned} F &= F_0(z + \psi(t)) - \psi'_t(t), \quad u = u_0(z + \psi(t), t), \\ v &= v_0(z + \psi(t), t), \quad w = w_0(z + \psi(t), t), \\ U &= U_0(z + \psi(t), t), \quad V = V_0(z + \psi(t), t), \end{aligned} \tag{6}$$

also yields the solution of system (5).

In particular, the stationary solution $F_0(z)$, $u_0(z)$, $v_0(z)$, $w_0(z)$, $U_0(z)$, $V_0(z)$, which is described by a system of ordinary differential equations (5) at α = const, β = const, γ = const, ε = const, δ = const, $\partial/\partial t = 0$, generates the respective nonstationary solution (6) of complete system (5). In this case, periodic functions $\psi(t)$ in (6) determine the periodic solution of system (5).

PHYSICAL INTERPRETATION OF THE SOLUTIONS AT $U = V = 0$

We consider the flows of a viscous incompressible fluid when the vector of the velocity of the fluid at the z axis is directed along this axis. Near the z axis, the radial components of velocity are small and can be expanded into the Taylor series in terms of the radial coordinates x and y. If we restrict ourselves to the main terms of expansion in terms of x and y in the components of velocity, it is possible to obtain representation (3) at $h_i = 0$, whence Eq. (4) follows at $U = V = 0$.

Any flows of a fluid that have two planes of symmetry admit representation (4) at $U = V = 0$ in the neighborhood of the intersection line of these planes (in the used notation, the intersection line of the planes determines the z axis). Among such flows are axisymmetric flows, the combinations of axisymmetric flows with rotation about the z axis (in particular, Karman flows), plane flows that are symmetric about a straight line, flows in straight impermeable and porous tubes with elliptic and rectangular cross sections, fluid jets that flow from holes of elliptic and rectangular forms, etc.

The specified property of the exact solutions of the Navier–Stokes equations of form (4) makes it possible to use them also for the approximate description of a considerably wider class of flows.

A scheme of the further study and classification of the flows described by system (5) is given in figure.

LAYERED FLOWS OF A FLUID ($F \equiv 0$)

Within the class of exact solutions (4), let us consider the flows of a fluid in which the velocity vector is parallel to the xy plane. Setting $V_3 = F = 0$ in (5), we obtain the overdetermined system

$$
\begin{gathered}
2(uv + w^2) = \alpha + \beta, \quad \frac{\partial u}{\partial t} = \nu\frac{\partial^2 u}{\partial z^2} + \gamma, \\
\frac{\partial v}{\partial t} = \nu\frac{\partial^2 v}{\partial z^2} + \gamma, \ \frac{\partial w}{\partial t} = \nu\frac{\partial^2 w}{\partial z^2} + \frac{\alpha - \beta}{2}, \\
\frac{\partial U}{\partial t} + wU + vV = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon, \\
\frac{\partial V}{\partial t} + uU - wV = \nu\frac{\partial^2 V}{\partial z^2} + \delta.
\end{gathered}
\tag{7}
$$

Not dwelling on the general analysis of the consistency of system (7), we give several physically informative examples of its nontrivial solubility.

Unidirectional flows. The "surplus" first equation of system (7) and the next three are identically satisfied if it is assumed that $u = v = w = \alpha = \beta = \gamma = 0$. Since, under the made assumptions, the velocity vector ceases to depend on the variables x and y, it is always possible to achieve its colinearity to the x axis by means of the rotation of the system of coordinates; i.e., without restricting the generality, it can be assumed that $V = \delta = 0$. As a result, we have

$$V_1 = U(t,z), \quad V_2 = V_3 = 0, \quad P = p_0 - \varepsilon x, \tag{8}$$

$$\frac{\partial U}{\partial t} = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon. \tag{9}$$

System of equations (8) and (9) contains a number of classical pressure and shift flows as its solutions.

1. *Plane Couette–Poiseuille flow* [17–19]. A viscous fluid contained between the planes $z = \pm h$ moves in a steady-state manner along the x axis under the effect of the given pressure gradient, which is equal to $-\varepsilon$, and the shear caused by the motion of the plane $z = h$ at constant velocity U_0. The solution of this problem that satisfies the no-slip conditions (i.e., the boundary conditions are added to (9): $U(h) = U_0$ and $U(-h) = 0$) is written as

$$U = \frac{\varepsilon h^2}{2\nu}\left(1 - \left(\frac{z}{h}\right)^2\right) + \frac{U_0}{2}\left(1 + \frac{z}{h}\right). \tag{10}$$

The general formulation of the problem of the flow of a fluid in cylindrical tubes of arbitrary cross sections can be found in many books (see, for example, [2, 4]).

2. *The first Stokes problem* [20]. At the initial time point, the plane $z = 0$, above which a viscous fluid is present, suddenly starts to move at a constant velocity in the direction of the x axis (in this case, in Eq. (9) it is assumed that $\varepsilon = 0$ and the following conditions are added: $U(t, 0) = U_0$, and $U(0, z) = 0$). The evolution of a disturbance that is brought into a medium in such a way occurs according to the self-similar law

$$U = U_0\left(1 - \frac{2}{\sqrt{\pi}}\int_0^{\xi}\exp(-\eta)^2 d\eta\right), \quad \xi = \frac{z}{2\sqrt{\nu t}}. \tag{11}$$

3. *The second Stokes problem* [20]. The plane $z = 0$ performs simple harmonic vibrations along the x axis with frequency ω and amplitude U_0/ω according to the law $x(t, 0) = (U_0/\omega)\sin(\omega t)$ (in this case, in Eq. (9) it is assumed that $\varepsilon = 0$ and the following conditions are added: $U(t, 0) = U_0\cos(\omega t)$, and $U(t, \infty) = 0$). As a result, in the fluid there also arises vibratory movement at the velocity

$$U = U_0\cos\left(\omega t - \sqrt{\frac{\omega}{2\nu}}z\right)\exp\left(-\sqrt{\frac{\omega}{2\nu}}z\right). \tag{12}$$

From formula (12), it follows that in moving away from the oscillating plane, the velocity exponentially rapidly decreases and, in this case, the vibration phase displacement occurs according to the linear law. Thus, the oscillating plane is drawn into the motion of the adjacent thin layer of the fluid.

Flows in a rotating fluid. Let a viscous fluid rotate as a solid at constant angular velocity Ω about the z axis. The analysis of over-determined system (7) shows that against the background of this rotation, a flow that is homogeneous with respect to variables x and y can occur at the velocity $(U, V, 0)$. In other words, assuming that $u = \Omega$, $v = -\Omega$, $\alpha = \beta = -\Omega^2/2$, and $w = \gamma = 0$, from (4) and (7) we derive

$$
\begin{gathered}
V_1 = U - \Omega y, \quad V_2 = V + \Omega x, \quad V_3 = 0, \\
P = p_0 - \varepsilon x - \delta y + \frac{\Omega^2}{4}(x^2 + y^2).
\end{gathered}
\tag{13}
$$

Here, the unknowns U and V satisfy the equations

$$\frac{\partial U}{\partial t} - \Omega V = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon, \quad \frac{\partial V}{\partial t} + \Omega U = \nu\frac{\partial^2 V}{\partial z^2} + \delta. \tag{14}$$

From (13) and (14), it can be seen that the centrifugal force that arises as a result of the rotation of the fluid is balanced by the part of the pressure gradient that is linear with respect to the variables x and y. The spatially homogeneous part of the pressure gradient is connected with the existence of the secondary flow with the velocity $(U, V, 0)$.

In the general case, the transformation of dependent variables

$$U = \bar{U}\cos(\Omega t) + \bar{V}\sin(\Omega t) + \frac{\delta}{\Omega},$$

$$V = -\bar{U}\sin(\Omega t) + \bar{V}\cos(\Omega t) - \frac{\varepsilon}{\Omega}$$

brings system (14) to two independent heat conduction equations

$$\frac{\partial \bar{U}}{\partial t} = \nu\frac{\partial^2 \bar{U}}{\partial z^2}, \quad \frac{\partial \bar{V}}{\partial t} = \nu\frac{\partial^2 \bar{V}}{\partial z^2}.$$

In the stationary case, the general solution of system (14) has the form

$$\begin{aligned} V &= (C_1\cosh(\omega z) + C_2\mathrm{sh}(\omega z))\cos(\omega z) \\ &+ (C_3\cosh(\omega z) + C_4\mathrm{sh}(\omega z))\sin(\omega z) + \frac{\varepsilon}{\Omega}, \\ U &= (C_4\cosh(\omega z) + C_3\mathrm{sh}(\omega z))\cos(\omega z) \\ &- (C_2\cosh(\omega z) + C_1\mathrm{sh}(\omega z))\sin(\omega z) + \frac{\delta}{\Omega}, \end{aligned} \tag{15}$$

where $\omega = \sqrt{|\Omega|/(2\nu)}$ is the constant with dimensionality reciprocal to length and C_i are the integration constants ($i = 1, 2, 3, 4$) that are determined, depending on problem formulation, by means of boundary and other additional conditions. Let us consider the specific examples.

1. *Ekman layer* [21]. At a large distance from the plane $z = 0$ that is rotated together with a fluid at angular velocity Ω, a constant wind $(U_0, V_0, 0)$ blows (i.e., V and U in formulas (15) should satisfy the following boundary conditions: $U(0) = V(0) = 0$, $U(\infty) = U_0$, $V(\infty) = V_0$). As a result, near the plane in the layer with a thickness of about ω^{-1}, there arises a flow of form (13) that is described by the relations

$$U = U_0 - (U_0\cos(\omega z) + V_0\sin(\omega z))\exp(-\omega z),$$

$$\delta = \Omega U_0,$$

$$V = V_0 - (V_0\cos(\omega z) - U_0\sin(\omega z))\exp(-\omega z),$$

$$\varepsilon = \Omega V_0.$$

2. *Enforced flow in a rotating layer* [22]. In the fluid layer, the solid boundaries of which $z = 0$ and $z = h$ rotate at constant angular velocity Ω about the z axis, a uniform axial pressure gradient $(-\varepsilon, -\delta, 0)$ is maintained (the part of the gradient that is linear with respect to x and y, as was stated above, balances a centrifugal force). As a result, there arises a shift flow with the velocity vector $(U, V, 0)$ that is calculated by formulas (15), in which the integration constants found from the boundary conditions $U(0) = U(h) = V(0) = V(h) = 0$ are as follows:

$$C_1 = -\frac{\varepsilon}{\Omega},$$

$$C_2 = \frac{(\delta\sin(\omega h) + \varepsilon\sinh(\omega h))(\cosh(\omega h) - \cos(\omega h))}{\Omega(\sin^2(\omega h) + \sinh^2(\omega h))},$$

$$C_4 = -\frac{\delta}{\Omega},$$

$$C_3 = \frac{(\delta\sinh(\omega h) - \varepsilon\sin(\omega h))(\cosh(\omega h) - \cos(\omega h))}{\Omega(\sin^2(\omega h) + \sinh^2(\omega h))}.$$

3. *Flow between planes rotating about different axes* [23]. The planes $z = 0$ and $z = h$ rotate at angular velocity Ω about the axes that pass through the points $x = 0$, $y = 0$ and $x = x_0$, $y = y_0$, respectively, inducing the motion of the fluid contained between them. The integration of the constants in (15) according to the boundary conditions $U(0) = V(0) = 0$, $U(h) = \Omega y_0$, $V(h) = -\Omega x_0$ (for simplicity, it is assumed that $\varepsilon = \delta = 0$) is as follows:

$$C_2 = \Omega\frac{y_0\sinh(\omega h)\cos(\omega h) + x_0\cosh(\omega h)\sin(\omega h)}{\sin^2(\omega h) + \sinh^2(\omega h)},$$

$$C_1 = 0,$$

$$C_3 = \Omega\frac{y_0\cosh(\omega h)\sin(\omega h) - x_0\sinh(\omega h)\cos(\omega h)}{\sin^2(\omega h) + \sinh^2(\omega h)},$$

$$C_4 = 0.$$

This solution can be used as the model of a flow in an orthogonal rheometer used for determining the rheological properties, for example, of non-Newtonian media.

A flow between disks that rotate at various velocities about different axes was considered within class (4) in [24].

The three solutions stated above give simple examples of swirl flows with a curvilinear axis of rotation (a line that consists of points in which the components of velocity vanish, $V_1 = V_2 = 0$).

PLANE FLOWS OF A FLUID

Let us consider the flows of a viscous fluid that have form (4) and are such that the pattern of motion remains unchanged in any plane that is parallel to the plane $y = 0$. Consequently, the hydrodynamic quantities in (4) do not depend on y, and the motion in the direction of this axis is absent. Thus, assuming in (4) that $u = v = V = \beta = \delta = \gamma = 0$, $w = -F_z/2$, we obtain

$$\begin{aligned} V_1 &= U - x\frac{\partial F}{\partial z}, \quad V_2 = 0, \quad V_3 = F, \\ P &= p_0 - \frac{\alpha}{2}x^2 - \varepsilon x - \frac{1}{2}F^2 - \int\frac{\partial F}{\partial t}dz + \nu\frac{\partial F}{\partial z}. \end{aligned} \tag{16}$$

Here, the unknowns F and U satisfy the equations

$$\frac{\partial^2 F}{\partial t \partial z} + F\frac{\partial^2 F}{\partial z^2} - \left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} - \alpha, \quad (17)$$

$$\frac{\partial U}{\partial t} + F\frac{\partial U}{\partial z} - U\frac{\partial F}{\partial z} = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon, \quad (18)$$

the first of which is isolated and the second is passive in the sense that it is necessary to search for its solution after Eq. (17) is solved. Among the solutions of Eq. (18), there are the trivial solution $U = \varepsilon = 0$ and the solution

$$U = n\frac{\partial F}{\partial z} + \frac{dn}{dt},$$

where F satisfies (17) and the time function n obeys the equation $d^2n/dt^2 = \alpha n + \varepsilon$.

Let us give some exact solutions of nonlinear equation (17):

$$F = a(t) + b(t)z, \quad b'_t - b^2 + \alpha = 0; \quad (19)$$

$$F = a'_t(t) + \frac{6\nu}{a(t) - z}, \quad \alpha = 0; \quad (20)$$

$$\begin{gathered} F = a(t)\exp(\lambda(t)z) + b(t)\exp(-\lambda(t)z) \\ + c(t)z + d(t), \\ \lambda'_t + c\lambda = 0, \quad c'_t - c^2 + 4\lambda^2 ab + \alpha = 0, \\ a'_t + (\lambda d - \nu\lambda^2 - 3c)a = 0, \\ b'_t - (\lambda d + \nu\lambda^2 + 3c)b = 0; \end{gathered} \quad (21)$$

$$\begin{gathered} F = \frac{a'_t(t)}{\omega(t)} + b(t)z + c(t)\cos(\omega(t)z - a(t)), \\ \omega'_t + b\omega = 0, \quad c'_t + (\nu\omega^2 - 3b)c = 0, \\ b'_t - b^2 - \omega^2 c^2 + \alpha = 0. \end{gathered} \quad (22)$$

In solutions (19) and (20), the function $a(t)$ can be chosen arbitrarily. Some other exact solutions of Eq. (17) will be considered further (see also [9, 25, 26]).

Remark. For solution (19), the respective equation (18) can be reduced to a linear thermal conduction equation with constant coefficients [9].

Flows near a stagnation point. A point at which velocity vanishes is called a stagnation or critical point. The field of the velocity of a plane potential flow in the neighborhood of a stagnation point is the special case of (16) and can be expressed as [5]

$$V_1 = -ax, \quad V_3 = az, \quad a = \text{const} \quad (V_2 = 0). \quad (23)$$

Formulas (16) and Eqs. (17) and (18) can be used to study the flow near a critical point, taking into account the forces of viscous friction. The interest in such studies is associated with the problems of a flow past solids and the boundary layer theory.

A flow between a critical point and a solid wall was first studied by Hiemenz [27] in the following formulation: a viscous fluid flows in a steady-state manner onto a solid plane that is perpendicular to the z axis; far from the plane, the distribution of the velocity of the incident flow has the form (23). This means that the stagnation point is located on the z axis and the incident flow is orthogonal to the plane ($U \equiv 0$). In this case, the function $F = F(z)$ satisfies Eq. (17), and the requirement of the no-slip of the fluid on the solid wall $z = 0$ and limiting relations (23) at infinity lead to the boundary conditions

$$F(0) = F'_z(0) = 0; \quad F'_z(\infty) = a. \quad (24)$$

The solution of boundary-value problem (17), (24) is not expressed in elementary functions. The approximate and numerical analysis of this problem [6] shows that a boundary layer that has a constant thickness of about $\sqrt{a/\nu}$ forms near the solid wall.

As an example of a flow with a stagnation point that is described by the solution of Eq. (17) ($U = 0$) expressed in elementary functions, let us consider a flow in front of a plane that moves in the direction of its normal line according to the law $z = Z(t)$. On the plane, the no-slip conditions are satisfied, and conditions (23) are satisfied far from the plane:

$$\begin{gathered} z = Z(t): \ F = Z'_t, \quad \frac{\partial F}{\partial z} = 0; \\ z \longrightarrow \infty: \ F \longrightarrow c(t)z. \end{gathered} \quad (25)$$

The solution of this problem can be expressed as (21) if the plane moves according to the law

$$Z = \frac{\nu}{2F_0}\sqrt{1+\tau}\ln(1+\tau), \quad \tau = \frac{6F_0^2}{\nu}t, \quad (26)$$

where F_0 is the initial velocity of the plane ($F_0 > 0$). Substituting expression (21) into boundary conditions (25) and taking into account relationship (26), we have $a(t) = d(t) = 0$. Ultimately, we find the velocity of the fluid

$$F = F_0^2\left(\frac{3z}{\nu(1+\tau)} + \exp\left(-\frac{3F_0 z}{\nu\sqrt{1+\tau}}\right)\right). \quad (27)$$

Solution (27) describes the decaying motion of the fluid in front of a decelerating plane.

A steady-state plane flow with a stagnation point on a plane that is inclined at a certain angle to an incident flow ($U \neq 0$) was studied in [28–30].

Exact solution (21) can be used for describing flows in a stagnation zone near a pervious plane that breaks down for a finite time. Collapsing flows are discussed in more detail in review [7] in connection with the problem of the symmetric deformation of a flat layer of a viscous fluid with two free boundaries. The particular solutions of this problem can be represented as (22). Various formulations and solutions of the problems of the motion of a fluid near a stagnation point are described in reviews [31–33].

Flows near a stretching or compressing plane. Let a plane that is perpendicular to the z axis stretch ($A < 0$) or compress ($A > 0$) according to the law

$$V_1 = -A(t)x, \tag{28}$$

where $A = A(t)$ is the given time function or a constant. Then, the motion of a fluid that can be described by expressions (16) arises near this plane. Boundary conditions (28) are not typical for hydrodynamics, but they can be used for the description of flows that occur near surfaces that are deformed, for example, during extrusion. Let us give two simple examples related to such motion.

1. *Steady-state flow near a stretching plane* [34]. Let us consider the flow of a fluid in a semi-infinite layer $z > 0$ that is induced by the stationary deformation of the plane $z = 0$ according to law (28). In this case, stationary equation (17) for the velocity $F = F(z)$ is completed with the following boundary conditions: $F(0) = 0$, $F'_z = -A$, and $F'_z(\infty) = 0$. The solution of the given problem is a special case of solution (21) and has the form

$$F = \sqrt{\nu A}\left(\exp\left(-\sqrt{\frac{A}{\nu}}z\right) - 1\right). \tag{29}$$

2. *A non-steady-state flow near a deforming plane.* The flow of a fluid in a semi-infinite layer $z > 0$ that is induced by the nonstationary deformation of the plane $z = 0$ according to law (28), where

$$A(t) = \frac{\nu\lambda^2 B}{B - \exp(-\nu\lambda^2 t)}, \tag{30}$$

is also described by solution (21) at $a(t) = c(t) = 0$:

$$F = \frac{A(t)}{\lambda}(e^{-\lambda z} - 1), \quad \lambda = \text{const} > 0. \tag{31}$$

It can be seen that the compression of the plane corresponds to any negative values of the constant B and $B > 1$, and, under conditions of the stretching of the plane ($0 < B < 1$), solution (31) becomes singular for a finite time.

Formulas (16) and Eqs. (17) and (18) can be used for the description of the flows of a viscous fluid in channels with compressing and stretching walls. The relevant problems at $U = 0$ (the x axis is parallel to the axis of the channel) are studied in [34, 35].

Flows in plane channels with pervious walls. A plane-parallel flow of a viscous fluid in a channel with walls of equal permeability and in the presence of an axial pressure gradient, within the class of exact solutions (16) ($U = 0$, the x axis is parallel to the axis of the channel), was first considered by Berman [36]. Berman obtained the asymptotic expansion of the solution at small Reynolds numbers constructed according to the width of the channel and the rate of the supply or withdrawal of the mass of the fluid through pervious lateral walls. Further numerical and asymptotic studies of flows in channels with pervious boundaries were carried out by a number of authors, asymptotic expansions for symmetric high-intensity injection and suction were constructed, and modes with the asymmetric flow of a fluid through boundaries were numerically studied [37]. A detailed description of the results of these studies and references to the published data can be found in book [38].

It should be noted that solutions (22) can be used for the description of some non-steady-state flows in a channel with porous walls, through which a fluid is supplied or withdrawn according to a special law.

ROTATION-SYMMETRIC KARMAN FLOWS

Rotation-symmetric flows are the nonlinear superposition of coaxial axisymmetric and purely rotary motions of a fluid. The determining relations for such flows are obtained as a result of the substitution of $w = U = V = \varepsilon = \delta = \gamma = 0$, $v = -u$, $\alpha = \beta$ into (4) and (5):

$$\begin{gathered} V_1 = -\frac{1}{2}\frac{\partial F}{\partial z}x - uy, \quad V_2 = ux - \frac{1}{2}\frac{\partial F}{\partial z}y, \quad V_3 = F, \\ P = p_0 - \frac{\alpha}{2}(x^2 + y^2) - \frac{1}{2}F^2 - \int\frac{\partial F}{\partial t}dz + \nu\frac{\partial F}{\partial z}, \end{gathered} \tag{32}$$

where the sought functions F and u satisfy the system of equations

$$\frac{\partial^2 F}{\partial t\partial z} + F\frac{\partial^2 F}{\partial z^2} - \frac{1}{2}\left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} - 2u^2 - 2\alpha, \tag{33}$$

$$\frac{\partial u}{\partial t} + F\frac{\partial u}{\partial z} - u\frac{\partial F}{\partial z} = \nu\frac{\partial^2 u}{\partial z^2}. \tag{34}$$

Using the class of solutions (32) written in cylindrical coordinates, Karman [39] studied the problem of the steady-state flow of a viscous fluid that is caused by the rotation of a plane at angular velocity Ω. The fluid adjacent to the plane is carried away in the radial direction by the centrifugal forces resulting from rotation. In this case, owing to the law of mass conservation, there arises a compensating flow that is directed to the plane along the rotation axis with the velocity tending to $V_3 = -0.886\sqrt{\nu\Omega}$ at $z \longrightarrow \infty$. The described principle of the organization of the directional flow of a fluid underlies the operation of centrifugal pumps, disk impellers, disk electrodes (used as sensors in electrochemistry), and many other technical devices. Radial and azimuthal motion in the Karman problem is almost completely confined to the layer with a thickness of about $\sqrt{\nu\Omega}$, that is adjacent to a rotating disk. The problem, which is reverse in a sense to the Karman problem, of motion near a stationary disk that is caused by the rotation of a fluid at infinity was studied in [40].

The distinctive features of flows such as (32) are the multiplicity of stationary solutions and their isolation from each other. These properties manifest themselves most vividly in the problem of the flow of a fluid between two planes that coaxially rotate at angular velocities $\Omega_1 \le \Omega_2 \ne 0$. Apparently, this problem was first formulated in [41]. After performing a qualitative analysis, Batchelor came to the conclusion that in almost the entire layer between the planes the fluid rotates as a solid and the boundary layers form near the planes. A different inference was made by Stewartson [42]. As a result of studying the motion of a fluid between stationary and rotating planes and during the rotation of the planes in different directions, Stewartson concluded that rotation is almost absent in the core and is localized in the boundary layers near the moving planes. Further studies showed the existence of both modes. Moreover, it was found [43, 44] that with an increase in the Reynolds number there, more and more new solutions with an increasingly more complex structure appear. A detailed analysis of the results of the study of the Batchelor–Stewartson problem can be found in [7], where other flows described by the Karman class are also discussed, including the problem of the evolution of a fluid layer with a free boundary on a rotating plane [45, 46].

By virtue of the nonlinearity of system of equations (33) and (34), the above and many other problems are mainly solved using approximate and numerical methods. Nevertheless, in studying the individual types of the motion of a fluid, the exact solutions of Eqs. (33) and (34) can be useful:

$$F = az^2 + b(t)z + c(t), \quad u = 0, \qquad b'_t - \frac{b^2}{2} + 2ac + 2\alpha = 0; \tag{35}$$

$$F = a'_t(t) + \frac{4\nu}{a(t) - z}, \quad u = 0, \quad \alpha = 0; \tag{36}$$

$$\begin{gathered} F = a(t) + b(t)z + c(t)\sin(\omega(t)z + \theta(t)), \\ u = A(t) + B(t)\sin(\omega(t)z + \theta(t)), \\ A'_t - bA = 0, \quad \omega'_t + b\omega = 0, \quad c'_t + (\nu\omega^2 - 2b)c = 0, \\ B = \pm\frac{\omega c}{2}, \quad A = \pm\frac{1}{2}(\theta'_t + \omega a), \\ b'_t - \frac{b^2}{2} + 2(A^2 - B^2 + \alpha) = 0. \end{gathered} \tag{37}$$

1. *Steady-state flow between radially compressing and stretching planes.* Let an impermeable plane that is specified by the equation $z = z_1$ radially compress according to the law $V_r = -Ar$ and a plane that is specified by the equation $z = z_2$ radially stretch at the same, in magnitude, velocity. Using representation (32), we find the boundary conditions for the vertical component of velocity: $F(z_1) = 0$, $F'_z(z_1) = 2A$; $F(z_2) = 0$, $F'_z(z_2) = -2A$. The motion of a fluid between the planes is described by an equation of form (35):

$$F = -\frac{2A}{h}(z - z_1)(z - z_2), \tag{38}$$

where $h = z_2 - z_1$ is the distance between the planes. The parameter α that is part of Eq. (33) is found by comparing (35) and (38).

It should be noted that the plane $z = h/2$ located in the middle between the planes is not stretchable, but permeable; therefore, Eq. (35) also describes a flow between permeable and compressing (stretching) planes.

Other stationary analogs of solution (35) are described in [47, 48].

2. *Symmetric spreading of a fluid layer under the action of a thermocapillary force.* The plane free boundary $z = h(t)$ of a fluid layer that is located on the solid support $z = 0$ is affected by the thermocapillary force $\nabla\tau = (\tau_0 x, \tau_0 y, 0)$ ($\tau_0 = \text{const} > 0$) that is radially directed and causes the spreading of the layer over the support with the stationary distribution of velocity

$$V_1 = \tau_0 xz, \quad V_2 = \tau_0 yz, \quad V_3 = F = -\frac{\tau_0}{\nu}z^2$$

and the nonstationary law of the movement of the free boundary

$$h(t) = \frac{\nu h_0}{\nu + \tau_0 h_0 t}.$$

Here, h_0 is the thickness of the layer at the initial time point $t = 0$.

In such a method of the initiation of motion, the radial pressure gradient that impedes the spreading does not arise ($\alpha = 0$) by virtue of the conditions on the free boundary [5] (the no-slip conditions are satisfied on the support):

$$\sigma_{zz}(t,h) = P_g(t), \quad \sigma_{xz}(t,h) = \frac{\partial\tau}{\partial x}, \qquad \sigma_{yz}(t,h) = \frac{\partial\tau}{\partial y}, \quad \frac{dh}{dt} = F(t,h). \tag{39}$$

Here, σ_{zz}, σ_{xz}, and σ_{yz} are the components of the tensor of viscous strains; $P_g(t)$ is the given pressure outside the layer that, without restricting the generality, can be considered equal to zero, since pressure is determined accurate to the arbitrary function $p_0(t)$.

Remark. The thermocapillary force originates as a consequence of the nonuniform dependence of the surface tension coefficient τ on the temperature T. This dependence is frequently assumed to be linear:

$$\tau = \bar{\tau} - \sigma T, \tag{40}$$

where the temperature coefficient of surface tension σ is, as a rule, a positive constant. The respective thermocapillary force

$$\nabla\tau = \left(-\sigma\frac{\partial T}{\partial x}, -\sigma\frac{\partial T}{\partial y}, 0\right) = (\tau_0 x, \tau_0 y, 0) \qquad (41)$$

can be created by the local heating of a spot of the free surface with radius R by a passive impurity with the homogeneous density of the volumetric heat generation Q that occurs, for example, as a consequence of chemical reactions (or heating by means of radiation). In this case, the temperature of the passive impurity and, together with it, the temperature of the free surface that satisfies the Poisson equation $\Delta T = -Q$, are distributed according to the law

$$T = T_0\left(1 - \frac{x^2 + y^2}{R^2}\right), \qquad (42)$$

where $T_0 = QR^2/4$ is the maximum temperature at the center of the spot (the temperature outside the spot is taken to be zero).

With the help of (39)–(42), it is easy to relate the positive constant τ_0 to the quantities σ, T_0, and R:

$$\tau_0 = \frac{2\sigma T_0}{R^2}.$$

3. *Flow near a radially compressing plane.* The plane $z = 0$ that is covered with a fluid (at $z > 0$) radially compresses at the velocity $V_r = -A(t)r$. The flow that arises can be described by (36) if we use the function $A(t) = [4(t_* - t)]^{-1}$, where $t \in [0, t_*)$. As a result, we obtain

$$F = \frac{4\nu z}{a(t)[a(t) + z]}, \quad a^2(t) = 8\nu(t_* - t). \qquad (43)$$

If the fluid moves away from the plane in the positive direction of the z axis ($a(t) > 0$), the normal component of the pressure gradient on its surface $P'_z = -8\nu^2/a^3$ is negative. It should be noted that in the stationary case, solution (36) goes over into the particular case of the known problem of the submerged jet of a fluid that was formulated and solved by Landau [5].

4. *Swirl flow of a viscous fluid near a solid boundary.* Let us consider a cylindrical glass in which a fluid near the free boundary is brought to rotation. In this case, poloidal circulation arises in the fluid, and a downward flow that entrains the particles of a passive impurity to the walls of the glass forms near the center, which is conventionally explained by the influence of centrifugal forces. As a consequence of the no-ship conditions at the bottom of the glass, such motion, apparently, will attenuate. If we restrict ourselves to the analysis of the situation only near the central part of the bottom of the glass, the problem can be solved within the considered class of solutions; namely, in order to solve (37), let us require the fulfillment of the no-slip conditions on the plane $z = 0$: $F(0, t) = F_z(0, t) = u(0, t) = 0$. As a result, we have

$$F = \frac{\nu}{3\omega^2}(\omega^3 + \omega_0^3)[\omega z - \sin(\omega z)],$$

$$u = \mp\frac{\nu}{6\omega}(\omega^3 + \omega_0^3)\sin(\omega z),$$

$$\omega'_t + \frac{\nu}{3}(\omega^3 + \omega_0^3) = 0,$$

where ω_0 is the arbitrary constant of integration.

The last equation is integrated in quadratures. Its general solution contains two (at $\omega_0 \neq 0$) monotonic branches tending to the common limit $\omega = -\omega_0$ at $t \longrightarrow \infty$. The motion in this case attenuates, but it occurs in an unusual way. When the function $\omega = \omega(t)$ that belongs to one of the branches crosses zero, poloidal circulation not only changes sign (F changes sign), but also is intensified (it is not difficult to select the initial data so that at the moment of reaching a maximum it is much higher than the initial value). This is in qualitative agreement with a real phenomenon in a glass when tea leaves are almost instantaneously collected at the center of the glass at some time point upon the termination of initial stirring. The fact of the presence of two different equilibrium states in the given problem with allowance for the effect of adhesion on the side walls of the glass, but without taking into account friction at the bottom, was proven earlier in [49] using another class of the exact solutions of the Navier–Stokes equations.

It should be noted that if it is assumed that viscosity is equal to zero, the solution of the problem under consideration such as (37) still exists, which is a rather rare example of the solutions of the Euler equations that satisfy the no-slip conditions.

Solutions (37) make it possible to study some special modes of flows between moving planes, on which the no-slip conditions are satisfied.

THREE-DIMENSIONAL FLOWS

Let us now turn to the general case of solutions with the linear dependence of the velocity of a flow on two coordinates (4) and (5). The main idea of the further analysis is to separate from system (5) a single isolated equation for the axial component of velocity $V_3 = F$.

Reduction of the first four equations of system (5) to a single equation. Let us first consider a special class of exact solutions that is described by a single equation. In the equations of system (5), we set

$$u = m\frac{\partial F}{\partial z} + A, \quad v = n\frac{\partial F}{\partial z} + B,$$
$$w = k\frac{\partial F}{\partial z} + C, \tag{44}$$

where m, n, k, A, B, and C are the sought functions of time t. We require that the first four equations of system (5) for F, u, v, and w be identical after the substitution of (44) into them. As a result, for the determination of the sought functions, we derive a nonlinear system that consists of one algebraic equation and six ordinary differential equations:

$$mn + k^2 = \frac{1}{4}, \tag{45}$$

$$\frac{A - m'_t}{m} = \frac{B - n'_t}{n} = \frac{C - k'_t}{k} = a(An + Bm + 2Ck), \tag{46}$$

$$\frac{\gamma - A'_t}{m} = \frac{\gamma - B'_t}{n} = \frac{\alpha - \beta - 2C'_t}{2k}$$
$$= -\alpha - \beta + 2AB + 2C^2. \tag{47}$$

This system contains seven equations for nine functions: six functions m, n, k, A, B, and C from (44) and three functions α, β, and γ from (5) (in this case, they are also considered to be sought). It is possible to show that the last equation in (46) is the corollary of the other three equations (45)–(47). Therefore, the three sought functions in system (45)–(47), generally speaking, can be chosen arbitrarily.

The first four equations of system (5) taking into account (45)–(47) are reduced to one equation

$$\frac{\partial^2 F}{\partial t \partial z} + F\frac{\partial^2 F}{\partial z^2} - \left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} + q\frac{\partial F}{\partial z} + p, \tag{48}$$

where the functions $p = p(t)$ and $q = q(t)$ are determined by the relations

$$p = \frac{\gamma - A'_t}{m}, \quad q = \frac{A - m'_t}{m}. \tag{49}$$

In constructing the solutions of system (45)–(47), it is necessary to discriminate two cases.

Solution of system (45)–(47) at $m = n$. In this case, the general solution of system (45)–(47) can be expressed as

$$m = n = \frac{1}{2}\sin\varphi, \quad k = \frac{1}{2}\cos\varphi,$$
$$A = B = \frac{1}{2}(q\sin\varphi + \varphi'_t\cos\varphi),$$
$$C = \frac{1}{2}(q\cos\varphi - \varphi'_t\sin\varphi),$$
$$\alpha = \frac{1}{4}q^2 + \frac{1}{4}(\varphi'_t)^2 - \frac{p}{2}(1 - \cos\varphi) + C'_t, \tag{50}$$
$$\beta = \frac{1}{4}q^2 + \frac{1}{4}(\varphi'_t)^2 - \frac{p}{2}(1 + \cos\varphi) - C'_t,$$
$$\gamma = \frac{p}{2}\sin\varphi + A'_t,$$

where $p(t)$, $q(t)$, and $\varphi(t)$ are the arbitrary functions. For the sake of convenience, the free functions p and q in (50) are chosen so that as a result of transformations (44) and (50), the first four equations of system (5) are reduced to one equation (48) with the same functions $p = p(t)$ and $q = q(t)$.

Thus, an important statement has been proven. Any solution of Eq. (48) for any functions $p = p(t)$ and $q = q(t)$ generates the exact solution of the Navier–Stokes equations (1). This solution is described by the function $F = F(z, t)$ and formulas (4), (44), and (50).

Solution of system (45)–(47) at $m \neq n$. In this case, the general solution of system (45)–(47) can be obtained as follows. The functions $m = m(t)$, $k = k(t)$, and $q = q(t)$ are set arbitrarily under the condition $m^2 + k^2 \neq 1/4$. The remaining functions that are present in system (45)–(47) and Eq. (48) are calculated sequentially by the formulas

$$n = \frac{1 - 4k^2}{4m}, \quad A = mq + m'_t, \quad B = nq + n'_t,$$
$$C = kq + k'_t, \quad p = \frac{A'_t - B'_t}{n - m}, \tag{51}$$
$$\alpha = AB + C^2 + C'_t - \frac{p}{2}(1 - 2k),$$
$$\beta = AB + C^2 - C'_t - \frac{p}{2}(1 + 2k), \quad \gamma = pm + A'_t.$$

In this case, the coefficient $p = p(t)$ in Eq. (48) is determined through the functions $m = m(t)$, $k = k(t)$, and $q = q(t)$ and their derivatives (not set arbitrarily, as was in the case of $m = n$). An attempt to specify directly the relationship $p = p(t)$ instead of specifying the function m (or k) leads to a nonlinear ordinary second-order differential equation for the function m (or k) with the arbitrary function $q(t)$.

Let us consider how the function $q = q(t)$ should be chosen so that the equality $p = 0$ is fulfilled. From the expression for p in (51), we have $A = B + s_0$, where s_0 is the arbitrary constant. From here, taking into account formulas (51) for the functions A, B, and n, we find the function q:

$$q = \frac{4s_0 m - 8(mm'_t + kk'_t)}{4(m^2 + k^2)} + \frac{m'_t}{m} \quad (\text{at } p = 0). \tag{52}$$

General property and exact solutions of Eq. (48). The general property of Eq. (48) is as follows: if $F_0(z, t)$ is its solution, the function

$$F = F_0(z + \psi(t), t) - \psi'_t(t), \tag{53}$$

where $\psi(t)$ is the arbitrary function, will also be the solution of Eq. (48).

At $q \neq 0$, Eq. (48) has the following exact solutions (the degenerate solution that is linear with respect to the coordinate z is omitted):

$$\begin{gathered} F = -6\nu k \tanh[k(z - \lambda t) + A] + \lambda, \\ p = 0, \quad q = 8\nu k^2; \end{gathered} \tag{54}$$

$$\begin{gathered} F = 6\nu k \tan[k(z - \lambda t) + A] + \lambda, \\ p = 0, \quad q = -8\nu k^2; \end{gathered} \tag{55}$$

$$\begin{gathered} F = \omega(t)z + \frac{\xi(t)}{\theta(t)}\sin[\theta(t)z + a], \\ \theta'_t = -\omega\theta, \quad \omega'_t = \omega^2 + q(t)\omega + p(t) + \xi^2, \\ \xi'_t = [2\omega - \nu\theta^2 + q(t)]\xi; \end{gathered} \tag{56}$$

$$\begin{gathered} F = \frac{a(t)}{\theta(t)}\exp[-\theta(t)z] + b(t) + c(t)z, \\ \theta'_t = -c\theta, \quad a'_t = (\nu\theta^2 + q + 2c + b\theta)a, \\ c'_t = c^2 + qc + p; \end{gathered} \tag{57}$$

$$\begin{gathered} F = \omega(t)z + \frac{\xi(t)}{\theta(t)}[Ae^{\theta(t)z} + Be^{-\theta(t)z}], \\ \theta'_t = -\omega\theta, \quad \omega'_t = \omega^2 + q(t)\omega + p(t) - 4AB\xi^2, \\ \xi'_t = [2\omega + \nu\theta^2 + q(t)]\xi; \end{gathered} \tag{58}$$

$$\begin{gathered} F = a'_t(t) + b(t)[z - a(t)] - \frac{6\nu}{z - a(t)}, \\ q = -4b, \quad p = b'_t + 3b^2. \end{gathered} \tag{59}$$

Here, A, B, k, and λ are arbitrary constants; equations (56)–(58) include functions that are described by the relevant systems of ordinary differential equations, and several functions can be chosen arbitrarily; and solution (59) includes two arbitrary functions $a(t)$ and $b(t)$. Some more exact solutions of Eq. (48) are given further.

It should be noted that property (53) makes it possible to generalize solutions (54), (55), and (58) by adding an arbitrary function to them.

Assuming in (56) that $\omega = 0$, $\theta = \text{const}$, and $\xi(t)$ is the periodic function, we obtain a solution that is periodic both in the time t and in the spatial coordinate z.

Let us consider the stationary case of solution (56). In formulas (50) and (56), we put

$$\begin{gathered} \varphi = 0, \quad \xi = -(a_1 + a_2), \quad q = \nu\theta^2 = 2a_1, \\ p = -\xi^2, \quad \theta = \sqrt{2a_1/\nu}. \end{gathered}$$

As a result, using expressions (4) and (44), we derive the solution

$$\begin{gathered} V_1 = a_1 x, \quad V_2 = [(a_1 + a_2)\cos(\theta z) - a_1]y, \\ V_3 = -\frac{a_1 + a_2}{\theta}\sin(\theta z), \end{gathered}$$

which describes the three-dimensional flow of the layer of a fluid between two plane elastic films (the position of the films is determined by the values $z = 0$ and $z = 2\pi/\theta$), the surfaces of which are stretched according to the law $V_1 = a_1 x$ and $V_2 = a_2 y$.

Reduction of the first four equations of system (5) to two equations. First case. Setting

$$\begin{gathered} u = a^2 G, \quad v = -b^2 G, \quad w = abG, \\ \alpha = \beta, \quad \gamma = 0, \end{gathered} \tag{60}$$

where a and b are arbitrary constants, we reduce the first four equations of system (5) to the isolated equation for the axial velocity F and to the additional equation for the function $G = G(z, t)$:

$$\frac{\partial^2 F}{\partial t \partial z} + F\frac{\partial^2 F}{\partial z^2} - \frac{1}{2}\left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} - 2\alpha, \tag{61}$$

$$\frac{\partial G}{\partial t} + F\frac{\partial G}{\partial z} - G\frac{\partial F}{\partial z} = \nu\frac{\partial^2 G}{\partial z^2}. \tag{62}$$

It can be seen that Eq. (61) is identical to Eq. (33) for the case of a nonswirling Karman flow at $u = 0$ and Eq. (62) is identical to Eq. (34) for the azimuthal component of velocity. Therefore, exact solutions (35) and (36) can be used to describe three-dimensional flows.

System of equations (61) and (62) has the exact solutions

$$F = az^2 + b(t)z + c(t), \quad G = A(t)z^2 + B(t) + C(t),$$

$$b_t' - \frac{b^2}{2} + 2ac + 2\alpha = 0, \quad A_t' + b(t)A - aB = 0,$$

$$B_t' + 2c(t)A - 2aC = 0, \quad C_t' + c(t)B - b(t)C = 2\nu A;$$

$$F = a_t'(t) + \frac{4\nu}{a(t)-z}, \quad \alpha = 0,$$

$$G = |z-a(t)|^{-3/2}[C_1\sin(\mu\ln|z-a(t)|) + C_2\cos(\mu\ln|z-a(t)|)], \quad \mu = \frac{\sqrt{7}}{2},$$

where C_1 and C_2 are arbitrary constants.

Let $F = F(z)$ be the stationary solution of Eq. (61). Then, Eq. (62) admits the nonstationary solution of exponential form

$$G(t,z) = e^{-\omega t}r(z), \quad Fr_z' - rF_z' = \nu r_{zz}'' + \omega r, \qquad (63)$$

where ω is an arbitrary constant, and the periodic solution

$$G(t,z) = \sin(\omega t)g(z) + \cos(\omega t)h(z), \qquad (64)$$

where the functions $g = g(z)$ and $h = h(z)$ are described by the system of ordinary differential equations

$$\begin{aligned} Fg_z' - gF_z' &= \nu g_{zz}'' + \omega h, \\ Fh_z' - hF_z' &= \nu h_{zz}'' - \omega g. \end{aligned} \qquad (65)$$

By virtue of the linearity of Eq. (62), solutions (64) obtained for different values $\omega = \omega_1, \ldots$, and $\omega = \omega_n$ (for the same $F = F(z)$) can be added.

Equation (62) for the stationary axial component of velocity $F(z)$ also has a more complex solution of the form $G(t, z) = e^{\mu t}[\sin(\omega t)g(z) + \cos(\omega t)h(z)]$.

Property (6) (the functions G, r, g, and h are transformed as u) makes it possible to generalize formulas (63)–(65).

Reduction of the first four equations of system (5) to two equations. Second case. In the equations of system (5), we put

$$u = \frac{1}{2}\sin\varphi\left(\frac{\partial F}{\partial z} + q\right) + a^2 G,$$

$$v = \frac{1}{2}\sin\varphi\left(\frac{\partial F}{\partial z} + q\right) - b^2 G,$$

$$w = \frac{1}{2}\cos\varphi\left(\frac{\partial F}{\partial z} + q\right) + abG,$$

$$\alpha = \frac{q^2}{4} - \frac{p}{2}(1-\cos\varphi) + \frac{q_t'}{2}\cos\varphi,$$

$$\beta = \frac{q^2}{4} - \frac{p}{2}(1+\cos\varphi) - \frac{q_t'}{2}\cos\varphi,$$

$$\gamma = \frac{1}{2}(p + q_t')\sin\varphi,$$

where $p = p(t)$ and $q = q(t)$ are the arbitrary functions, a and b are the arbitrary constants, $G = G(z, t)$ is the sought function, and φ is the constant that is determined from the transcendental equation

$$(a^2 - b^2)\sin\varphi + 2ab\cos\varphi = 0.$$

As a result, the first four equations of system (5) are reduced to two equations

$$\frac{\partial^2 F}{\partial t\partial z} + F\frac{\partial^2 F}{\partial z^2} - \left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} + q\frac{\partial F}{\partial z} + p, \qquad (66)$$

$$\frac{\partial G}{\partial t} + F\frac{\partial G}{\partial z} - G\frac{\partial F}{\partial z} = \nu\frac{\partial^2 G}{\partial z^2}. \qquad (67)$$

Equation (66) is identical to Eq. (48) and has exact solutions (54)–(59). The stationary variant of Eq. (66) was considered in [50].

Equation (66) has solution (59). At $b(t) = 0$, by means of the transformation

$$G = \zeta^{-3}\Psi(\zeta,t), \quad \zeta = z - a(t),$$

Eq. (67) is reduced to the classical heat conduction equation

$$\frac{\partial\Psi}{\partial t} = \nu\frac{\partial^2\Psi}{\partial\zeta^2}.$$

Let $F = F(z, t)$ be the solution of Eq. (66). Then, Eq. (67) has the solution

$$G = A\frac{\partial F}{\partial z} + A_t' + Aq, \qquad (68)$$

where the function $A = A(t)$ satisfies the ordinary differential equation

$$A_{tt}'' + qA_t' + (p + q_t')A = 0. \qquad (69)$$

Problems of the nonlinear stability of the solutions of system (66)–(67). Let us now consider in more detail problems with a stationary axial component of velocity, which corresponds to $F = F(z)$, $p = \text{const}$, and $q = \text{const}$. In this case, the solution of Eq. (69) depends on the sign of the discriminant $\Delta = q^2 - 4p$:

$$A(t) = \begin{cases} \exp\left(-\frac{qt}{2}\right)\left[C_1\exp\left(\frac{t\sqrt{\Delta}}{2}\right) + C_2\exp\left(-\frac{t\sqrt{\Delta}}{2}\right)\right] \\ \text{at } \Delta > 0, \\ \exp\left(-\frac{qt}{2}\right)\left[C_1\sin\left(\frac{t\sqrt{|\Delta|}}{2}\right) + C_2\cos\left(-\frac{t\sqrt{|\Delta|}}{2}\right)\right] \\ \text{at } \Delta < 0, \\ \exp\left(-\frac{qt}{2}\right)(C_1 t + C_2) \\ \text{at } \Delta = 0, \end{cases} \qquad (70)$$

where C_1 and C_2 are arbitrary constants.

At $q < 0$ (at any p) and $p < 0$ (at any q), solutions (68) and (70) exponentially increase at $t \longrightarrow \infty$. Therefore, the specified values of the parameters determine the region of the absolute instability of system (66)–(67) for any restricted stationary profile of the axial component of velocity $F(z)$ (distinct from a constant). The point $p = q = 0$ also falls into the region of the instability of system (66)–(67).

At $q = 0$ and $p > 0$, solution (70) and, consequently, solution (68), are periodic. The joint fulfillment of the inequalities $p \geq 0$ and $q \geq 0$ ($|p| + |q| \neq 0$) determines the region of the conditional stability of the solutions under consideration.

It should be emphasized that here we are dealing with *nonlinear instability*, since all given solutions are exact (rather than linearized, as is the case of the theory of linear stability).

All above derivations on stability and instability and formulas (68)–(70) remain valid for any nonstationary solutions $F = F(t, z)$ of Eq. (66) at p = const and q = const.

Remark 1. Similarly, using Eq. (69), it is possible to test the nonstationary solutions of Eq. (66) for stability at variables $p = p(t)$ and $q = q(t)$.

Remark 2. Let $F = F(t, z)$ be the stationary solution of Eq. (66). Then, Eq. (67) admits a nonstationary exact solution of exponential type (63) and a periodic solution of form (64)–(65).

Some other exact solutions of system of equations (66) and (67) are given further.

Three-dimensional flow at $u = v = w = 0$. Setting in (4) and (5) $u = v = w = 0$, $\alpha = \beta$, $\gamma = 0$ we obtain a rather simple three-dimensional flow that has the following form:

$$V_1 = U - \frac{x}{2}\frac{\partial F}{\partial z}, \quad V_2 = V - \frac{y}{2}\frac{\partial F}{\partial z}, \quad V_3 = F,$$
$$P = p_0 - \frac{\alpha}{2}(x^2 + y^2) - \varepsilon x - \delta y - \frac{1}{2}F^2 - \int \frac{\partial F}{\partial t} dz + \nu \frac{\partial F}{\partial z}, \tag{71}$$

where the functions U, V, and F are described by the equations

$$\frac{\partial^2 F}{\partial t \partial z} + F\frac{\partial^2 F}{\partial z^2} - \frac{1}{2}\left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} - 2\alpha,$$
$$\frac{\partial U}{\partial t} + F\frac{\partial U}{\partial z} - \frac{U}{2}\frac{\partial F}{\partial z} = \nu\frac{\partial^2 U}{\partial z^2} + \varepsilon, \tag{72}$$
$$\frac{\partial V}{\partial t} + F\frac{\partial V}{\partial z} - \frac{V}{2}\frac{\partial F}{\partial z} = \nu\frac{\partial^2 V}{\partial z^2} + \delta.$$

It will be recalled that the quantities p_0, α, ε, and δ can depend on time. Note that even at $V = \delta = 0$, the flow under consideration remains three-dimensional.

The last two similar equations in (72) do not depend on each other. Therefore, we can restrict ourselves to the analysis of the first two equations in (72).

Let $F = F(z, t)$ be some solution of the first equation in (72). Then, the second equation in (72) has the solution

$$U = s(t)\frac{\partial F}{\partial z} + 2s'_t(t), \tag{73}$$

where the function $s = s(t)$ is described by the ordinary differential equation

$$s''_{tt} + \alpha(t)s - \frac{\varepsilon(t)}{2} = 0. \tag{74}$$

The first equation in (72) is identical to Eq. (33) (at $u = 0$) and with Eq. (61). Therefore, exact solutions (35) and (36) are the solutions of the first equation in (72). Formulas (73) and (74) make it possible to construct the respective solutions of the second equation in (72).

Let $F = F(z)$ be the stationary solution of the first equation in (71). Then, the second equation in (71) admits the nonstationary exact solution of exponential type (63) and the periodic solution of form (64)–(65).

LINEAR AND NONLINEAR TRANSFORMATIONS OF THE EQUATION FOR THE AXIAL COMPONENT OF VELOCITY *F*. EXACT SOLUTIONS

Linear transformations of the equation for the component of velocity F. Let us consider the equation

$$\frac{\partial^2 F}{\partial t \partial z} + F\frac{\partial^2 F}{\partial z^2} - m\left(\frac{\partial F}{\partial z}\right)^2 = \nu\frac{\partial^3 F}{\partial z^3} + q(t)\frac{\partial F}{\partial z} + p(t), \tag{75}$$

which is identical to (48) and (66) at $m = 1$ (at $q(t) = 0$, see also (17)) and to Eq. (61) at $m = 1/2$ and $q(t) = 0$ (see also (33) at $u = 0$). At $m = 1$, both functions $p(t)$ and $q(t)$ can be chosen arbitrarily.

The general property of Eq. (75) is as follows: if $F_0(z, t)$ is its solution, function (53), where $\psi(t)$ is an arbitrary function, will also be the solution of Eq. (75).

The linear transformation with respect to the sought functions

$$F = a(t)f(\tau, \xi) + b(t)z + c(t),$$
$$\xi = \lambda(t)z + \sigma(t), \quad \tau = \int \lambda^2(t)dt + C_0, \tag{76}$$

where $a = a(t)$, $b = b(t)$, $c = c(t)$, $\lambda = \lambda(t)$, and $\sigma = \sigma(t)$ are arbitrary functions; brings Eq. (75) to the form

$$\frac{\partial^2 f}{\partial \tau \partial \xi} + [\tilde{a}f + \tilde{b}\xi + \tilde{c}]\frac{\partial^2 f}{\partial \xi^2} - m\tilde{a}\left(\frac{\partial f}{\partial \xi}\right)^2 = \nu\frac{\partial^3 f}{\partial \xi^3} + \tilde{q}\frac{\partial f}{\partial \xi} + \tilde{p}, \tag{77}$$

where the following designations are used:

$$\tilde{a}(\tau) = \frac{a}{\lambda}, \quad \tilde{b}(\tau) = \frac{1}{\lambda^3}(b\lambda + \lambda'_t),$$

$$\tilde{c}(\tau) = \frac{1}{\lambda^3}(c\lambda^2 - b\lambda\sigma + \lambda\sigma'_t - \sigma\lambda'_t),$$

$$\tilde{q}(\tau) = \frac{1}{a\lambda^3}\left[aq\lambda + 2mab\lambda - \frac{d(a\lambda)}{dt}\right], \quad (78)$$

$$\tilde{p}(\tau) = \frac{1}{a\lambda^3}(p + bq + mb^2 - b'_t).$$

The argument of the functions in the left-hand sides of the equalities is τ, and the argument of the functions in the right-hand sides of the equalities is t; the variables τ and t are related by the last equation in (76).

The presence of a large number (from five to seven) of arbitrary functions in (75)–(78) makes it possible to construct different exact solutions of Eq. (75).

Example 1. Setting in (77) $f = 1, f = \xi^{-1}, f = e^{\xi}, f = \sin\xi$ and choosing arbitrary functions and the parameter m, we can obtain all solutions (19)–(22), (35)–(36), and (56)–(59).

Example 2. Equation (77) has the stationary solutions

$$f(\xi) = \frac{1}{1 \pm e^{\xi}}, \quad (79)$$

which, with allowance for (79), lead to new exact solutions of the original equation (75) that have the following form:

$$F(z,t) = \frac{a(t)}{1 \pm \exp[\lambda(t)z + \sigma(t)]} + b(t)z + c(t). \quad (80)$$

The following coefficients in Eq. (77) correspond to solution (79):

$$\tilde{a} = \frac{6\nu}{2-m}, \quad \tilde{b} = 0, \quad \tilde{c} = -\frac{3\nu}{2-m},$$

$$\tilde{p} = 0, \quad \tilde{q} = \frac{1+m}{2-m}\nu.$$

Substituting these expressions into (78), we can derive a system of ordinary differential equations for determining the functional coefficients of solution (80) (at $m = 1$, two functions in this system can be chosen arbitrarily).

Example 3. Setting

$$\tilde{a} = C_1, \ \tilde{b} = C_2, \ \tilde{c} = C_3, \ \tilde{p} = C_4, \ \tilde{q} = C_5, \quad (81)$$

where C_n are arbitrary constants, from (77) we derive an ordinary differential equation for the function $f = f(\xi)$. In this case, relations (78) under condition (81) are a system of ordinary differential equations for the functional coefficients of transformation (76). At $m = 1$ in Eq. (75) and transformation (76) subject to limitations (78)–(81), two functions can be specified arbitrarily. The stationary solution $f = f(\xi)$ of Eq. (77) generates a nonstationary solution such as the generalized traveling wave (76) of the original equation (75).

Example 4. We now set

$$\tilde{a} = C_1\tau^{-\frac{k+1}{2}}, \quad \tilde{b} = C_2\tau^{-1}, \quad \tilde{c} = C_3\tau^{-\frac{1}{2}},$$
$$\tilde{q} = C_4\tau^{-1}, \quad \tilde{p} = C_5\tau^{\frac{k-3}{2}}, \quad \tau = \int\lambda^2(t)dt + C_0, \quad (82)$$

where k and C_n are arbitrary constants. In this case, Eq. (82) admits the self-similar solution

$$f = \tau^{k/2}h(\zeta), \quad \zeta = \xi\tau^{-1/2},$$

where the function $h = h(\zeta)$ satisfies the ordinary differential equation

$$\left(\frac{k-1}{2} - C_4\right)h'_\zeta + \left[C_1 h + \left(C_2 - \frac{1}{2}\right)\zeta + C_3\right]h''_{\zeta\zeta}$$
$$- mC_1(h'_\zeta)^2 = \nu h'''_{\zeta\zeta\zeta} + C_5. \quad (83)$$

Substituting (82) into (78), we obtain a system of integrodifferential equations for determining the functional parameters of the original equation (75) and transformation (76). It should be noted that the replacement $\lambda = \sqrt{\varphi'_t}$ with allowance for $\tau = \varphi(t) + C_0$ makes it possible to come to a standard system of ordinary differential equations. The specified solution is not self-similar and contains two arbitrary functions.

Example 5. Now, $\tilde{a} = 1$ and $\tilde{b} = \tilde{c} = 0$. From the first three relations in (78), we derive

$$\lambda = a, \quad b = -a'_t/a, \quad c = -\sigma'_t/a, \quad (84)$$

and the last two relations in (78) become

$$\tilde{q}(\tau) = \frac{1}{a^2}\left[q(t) - 2(m+1)\frac{a'_t}{a}\right], \quad \tau = \int a^2(t)dt + C_0,$$
$$\tilde{p}(\tau) = \frac{1}{a^4}\left[p(t) - \frac{a'_t}{a}q(t) + (m-1)\left(\frac{a'_t}{a}\right)^2 + \frac{a'_{tt}}{a}\right]. \quad (85)$$

Thus, transformation (76), into which it is necessary to substitute relations (84), brings Eq. (75) to the equation of similar form (77) with coefficients (85). Formulas (84) and (85) contain two arbitrary functions $a = a(t)$ and $\sigma = \sigma(t)$; in addition, at $m = 1$ in Eq. (75) the functions $p(t)$ and $q(t)$ can also be arbitrary. By choosing the functions $a(t)$, $\sigma(t)$, $p(t)$, and $q(t)$ in an appropriate manner, it is possible to construct and reproduce the exact solutions of the original equation.

In particular, at

$$q(t) = 2(m+1)\frac{a'_t}{a}, \quad p(t) = (m+3)\left(\frac{a'_t}{a}\right)^2 - \frac{a''_{tt}}{a}, \quad (86)$$

where $a = a(t)$ is an arbitrary function, formulas (85) yield the zero values $\tilde{q} = \tilde{p} = 0$ and Eq. (77) is reduced to the autonomous form

$$\frac{\partial^2 f}{\partial\tau\partial\xi} + f\frac{\partial^2 f}{\partial\xi^2} - m\left(\frac{\partial f}{\partial\xi}\right)^2 = \nu\frac{\partial^3 f}{\partial\xi^3}.$$

Any stationary solution of this equation $f = f(\xi)$ generates the nonstationary solution (76), (84) of Eq. (75) with coefficients (86).

Linear transformations of the system of equations (75) and (67). Let us consider system of equations (75) and (67), which generalizes system of equations (66) and (67) and at $m = 1$ and $q = 0$ goes over into system of equations (61) and (62). We make the linear transformation of the sought functions that consists of transformation (76) and the transformation

$$G = s(t)g(\tau,\xi), \quad \xi = \lambda(t)z + \sigma(t), \qquad \tau = \int\lambda^2(t)dt + C_0. \tag{87}$$

As a result, we derive Eq. (77) for the function f and the following equation for the function g:

$$\frac{\partial g}{\partial\tau} + (\tilde{a}f + \tilde{b}\xi + \tilde{c})\frac{\partial g}{\partial\xi} - \tilde{a}g\frac{\partial f}{\partial\xi} + \tilde{s}g = \nu\frac{\partial^2 g}{\partial\xi^2}. \tag{88}$$

Here, the expressions for the functional coefficients $\tilde{a}$, $\tilde{b}$, and $\tilde{c}$ are given in (78), and $\tilde{s}$ is determined by the formula

$$\tilde{s} = \frac{1}{s\lambda^2}(s'_t - bs). \tag{89}$$

Equations (77), (88) and relations (78), (89) make it possible to construct the exact solutions of system of equations (75) and (67) or (66) and (67) with the use of formulas (76) and (87).

Example 6. Imposing conditions (81) and $\tilde{s} = C_6$ on the coefficients, from (77) and (88) we derive a system of ordinary differential equations for determining the functions $f(\xi)$ and $g(\xi)$. Relations (78) and (89) under conditions (81) and $\tilde{s} = C_6$ yield a system of ordinary differential equations that makes it possible to find the functional coefficients in formulas (76) and (87). At $m = 1$, a solution such as a generalized traveling wave that is obtained in this way contains two arbitrary functions.

Example 7. We now add the relationship $s = C_6\tau^{-1}$ to (82). In this case, system of equations (77) and (88) admits the self-similar solution

$$f = t^{k/2}h(\zeta), \quad g = \tau^{\mu}r(\zeta), \quad \zeta = \xi\tau^{-1/2},$$

where μ is an arbitrary constant, the function $h(\zeta)$ satisfies Eq. (83), and the function $r(\zeta)$ satisfies the equation

$$\left[C_1 h + \left(C_2 - \frac{1}{2}\right)\zeta + C_3\right]r'_\zeta - C_1 r h'_\zeta + (C_6 + \mu)r = \nu r''_{\zeta\zeta}.$$

Relations (78) and (89) together with conditions (82) and $\tilde{s} = C_6\tau^{-1}$ yield equations for determining the functional coefficients in solution (76)–(87) (note that this is not a self-similar solution).

Nonlinear Krokko transformation. Let us introduce the following designations:

$$\eta = \frac{\partial F}{\partial z}, \quad \Phi = \frac{\partial^2 F}{\partial z^2}. \tag{90}$$

We rearrange the term mF_z^2 (hereinafter, an abridged notation of derivatives is used) to the right-hand side of Eq. (75), divide both parts of the obtained expression by $F_{zz} = \Phi$, and differentiate it with respect to z. Taking into account (90), we have

$$\frac{\Phi_t}{\Phi} - \frac{F_{zt}\Phi_z}{\Phi^2} + \eta = \frac{\partial}{\partial z}\frac{\nu\Phi_z + m\eta^2 + q(t)\eta + p(t)}{\Phi}. \tag{91}$$

Let us pass in (91) from the old variables t, z, and $F = F(z, t)$ to the new variables t, η, and $\Phi = \Phi(t, \eta)$, where η and Φ are determined by formulas (90). In converting from the old variables to the new variables, derivatives are transformed as follows:

$$\frac{\partial}{\partial z} = \frac{\partial\eta}{\partial z}\frac{\partial}{\partial\eta} = F_{zz}\frac{\partial}{\partial\eta} = \Phi\frac{\partial}{\partial\eta},$$
$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t} + \frac{\partial\eta}{\partial t}\frac{\partial}{\partial\eta} = \frac{\partial}{\partial t} + F_{tz}\frac{\partial}{\partial\eta}.$$

As a result, (91) is reduced to the second-order equation

$$\frac{\eta}{\Phi} - \frac{\partial}{\partial t}\frac{1}{\Phi} = \frac{\partial}{\partial\eta}\left(\frac{m\eta^2 + q\eta + p}{\Phi}\right) + \nu\frac{\partial^2\Phi}{\partial\eta^2}$$

or

$$\frac{\partial\Phi}{\partial t} + (m\eta^2 + q\eta + p)\frac{\partial\Phi}{\partial\eta} = [(2m-1)\eta + q]\Phi + \nu\Phi^2\frac{\partial^2\Phi}{\partial\eta^2}. \tag{92}$$

Note that, in the degenerate case (a nonviscous fluid, $\nu = 0$), the original nonlinear second-order equation (75) is reduced to the linear first-order equation (92), which can be completely integrated by the method of characteristics.

If the solution of the original equation (75) is known, formulas (90) yield the parametric form of the solution of Eq. (92).

Let $\Phi = \Phi(\eta, t)$ be some solution of Eq. (92). Then, the respective solution of the original equation (75) can also be expressed in the parametric form

$$z = \int\frac{ds}{\Phi(s,t)} + \psi(t), \quad F = \int\frac{sds}{\Phi(s,t)} - \psi'_t(t),$$

where $\psi(t)$ is an arbitrary function (in integrating, t is regarded as a parameter).

Let us consider again system of equations (75) and (67), which generalizes system of equations (66) and (67) and, at $m = 1$ and $q = 0$, goes over into system of equations (61) and (62). Assuming that $F_{zz} \neq 0$, we convert from the variables x and t to the new variables t and η according to (90). Equation (75) is transformed to Eq. (92), and Eq. (67) is transformed to the equation

$$\frac{\partial G}{\partial t} + (m\eta^2 + q\eta + p)\frac{\partial G}{\partial \eta} - \eta G = \nu\Phi^2\frac{\partial^2 G}{\partial \eta^2}. \quad (93)$$

Further, we assumed that $p = \text{const}$ and $q = \text{const}$. In this case, Eq. (75) has an exact solution such as a traveling wave $F = F(kz - \lambda t)$. The stationary solution of Eq. (92) $\Phi = \Phi(\eta)$ corresponds to this solution. Equation (93) has the respective nonstationary solution with the separable variables $G(t, \eta) = e^{\mu t}g(\eta)$ where the function $g(\eta)$ is described by the ordinary differential equation

$$(m\eta^2 + q\eta + p)g'_\eta + (\mu - \eta)g = \nu\Phi^2(\eta)g''_{\eta\eta}.$$

At $m = 1/2$ and $q = 0$, the exact solution $F = az^2 + b(t)z + c(t)$ of Eq. (75) corresponds to $\Phi = 2a = \text{const}$. In this case, Eq. (93) also admits the nonstationary solution with the separable variables $G(t, \eta) = e^{\mu t}g(\eta)$.

SOME EXACT SOLUTIONS OF THE COMPLETE SYSTEM OF EQUATIONS (5)

Solutions as a traveling wave. The following statement is valid. Let the functions $F_0(z)$, $u_0(z)$, $v_0(z)$, $w_0(z)$, $U_0(z)$, $V_0(z)$ determine the general solution of stationary system (5) at $\alpha = \text{const}$, $\beta = \text{const}$, $\gamma = \text{const}$, $\varepsilon = \text{const}$, $\delta = \text{const}$, and $\partial/\partial t = 0$. Then, all solutions such as a traveling wave of nonstationary system (5) at $\alpha = \text{const}$, $\beta = \text{const}$, $\gamma = \text{const}$, $\varepsilon = \text{const}$, and $\delta = \text{const}$ are described by formulas

$$\begin{gathered} F = F_0(z - \lambda t) + \lambda, \quad u = u_0(z - \lambda t), \\ v = v_0(z - \lambda t), \quad w = w_0(z - \lambda t), \\ U = U_0(z - \lambda t), \quad V = V_0(z - \lambda t), \end{gathered} \quad (94)$$

where λ is an arbitrary constant.

From formula (94), it follows that if the function $F_0(z)$ that determines the stationary solution tends to zero at $z \longrightarrow \infty$, the respective traveling wave that propagates to the right at the velocity $\lambda > 0$ is described by the function F, which tends to λ at $z \longrightarrow \infty$.

Self-similar solutions. For the determining functions

$$\begin{gathered} \alpha = \frac{\alpha_0}{t^2}, \quad \beta = \frac{\beta_0}{t^2}, \quad \gamma = \frac{\gamma_0}{t^2}, \\ \varepsilon = \frac{\varepsilon_0}{t^2}, \quad \delta = \frac{\delta_0}{t^2}, \end{gathered} \quad (95)$$

where α_0, β_0, γ_0, ε_0, and δ_0 are arbitrary constants; system of equations (5) admits the self-similar solution

$$\begin{gathered} F = \frac{1}{\sqrt{t}}f(\zeta), \quad u = \frac{1}{t}g_1(\zeta), \quad v = \frac{1}{t}g_2(\zeta), \\ w = \frac{1}{t}g_3(\zeta), \; U = \frac{1}{t}h_1(\zeta), \; V = \frac{1}{t}h_2(\zeta), \; \zeta = \frac{z}{\sqrt{t}}, \end{gathered} \quad (96)$$

where the functions $f(\zeta)$ and $g_n(\zeta)$ are described by the system of ordinary differential equations

$$\begin{gathered} -f' - \left(\frac{\zeta}{2} - f\right)f'' - \frac{1}{2}(f')^2 \\ = -(\alpha_0 + \beta_0) + \nu f''' + 2(g_1 g_2 + g_3^2), \\ \left(f - \frac{\zeta}{2}\right)g'_1 - (f' + 1)g_1 = \nu g''_1 + \gamma_0, \\ \left(f - \frac{\zeta}{2}\right)g'_2 - (f' + 1)g_2 = \nu g''_2 + \gamma_0, \\ \left(f - \frac{\zeta}{2}\right)g'_3 - (f' + 1)g_3 = \nu g''_3 + \frac{\alpha_0 - \beta_0}{2}, \\ \left(f - \frac{\zeta}{2}\right)h'_1 - \left(\frac{f'}{2} - g_3 + 1\right)h_1 + g_2 h_2 = \nu h''_1 + \varepsilon_0, \\ \left(f - \frac{\zeta}{2}\right)h'_2 - \left(\frac{f'}{2} + g_3 + 1\right)h_2 + g_1 h_1 = \nu h''_2 + \delta_0. \end{gathered} \quad (97)$$

At $\gamma_0 = \varepsilon_0 = \delta_0 = 0$, solution (95)–(97) was obtained in [16].

It should be noted that by making the shift in time $t \longrightarrow t + C$ in (95) and (96), singularity in the solution at $t = 0$ can be avoided.

Solutions as the generalized traveling wave and the generalized self-similar solutions. Using the linear transformation of the sought functions

$$\begin{gathered} F = a(t)f(\tau,\xi) + b(t)x + c(t), \; \xi = \lambda(t)z + \sigma(t), \\ \tau = \int \lambda^2(t)dt, \; u = s(t)\varphi_1(\tau,\xi), \\ v = s(t)\varphi_2(\tau,\xi), \; w = s(t)\varphi_3(\tau,\xi), \\ U = r(t)\psi_1(\tau,\xi), \; V = r(t)\psi_2(\tau,\xi), \end{gathered}$$

by means of the appropriate choice of the functional parameters $a(t)$, $b(t)$, $c(t)$, $\lambda(t)$, $\sigma(t)$, $s(t)$, and $r(t)$, system (5) can be reduced to a form that admits solutions such as a traveling wave and self-similar solutions in the variables ξ and τ. In returning to the original variables x and t, these solutions are transformed to solutions such as a generalized traveling wave and generalized self-similar solutions. A detailed procedure for deriving these solutions, due to its awkwardness, is not given in this paper (this procedure was described in detail in the previous section for the case of one equation (75) and systems of two equations (75) and (67)).

Solution containing ten arbitrary functions. A system of the first four equations in (5) admits the exact solution

$$F = a(t)z + b(t), \quad u = u(t),$$
$$v = u(t) + C\exp\left(\int a(t)dt\right), \quad w = w(t), \tag{98}$$

where $a = a(t)$, $b = b(t)$, $u = u(t)$, and $w = w(t)$ are the arbitrary functions through which the free functions present in (4) are expressed:

$$\alpha = \frac{1}{4}a^2 - \frac{1}{2}a_t' + uv + w^2 - aw + w_t',$$
$$\beta = \frac{1}{4}a^2 - \frac{1}{2}a_t' + uv + w^2 + aw - w_t', \tag{99}$$
$$\gamma = u_t' - au.$$

Let us substitute (98) into the last two equations of system (5) and transform the dependent variables by the formulas

$$U = \varphi_1(t)R(z,t) + \varphi_2(t)S(z,t) + \varphi_0(t),$$
$$V = \psi_1(t)R(z,t) + \psi_2(t)S(z,t) + \psi_0(t), \tag{100}$$

where the pairs of functions $\varphi_1 = \varphi_1(t)$, $\psi_1 = \psi_1(t)$ and $\varphi_2 = \varphi_2(t)$, $\psi_2 = \psi_2(t)$ are the linearly independent (fundamental) solutions of a homogeneous system of the linear differential equations

$$\varphi_t' = \left(\frac{a}{2} - w\right)\varphi - v\psi, \quad \psi_t' = \left(\frac{a}{2} + w\right)\psi - u\varphi, \tag{101}$$

and the functions $\varphi_0 = \varphi_0(t)$ and $\psi_0 = \psi_0(t)$ are the particular solutions of an inhomogeneous system of the linear ordinary differential equations

$$\frac{d\varphi_0}{dt} = \left(\frac{a}{2} - w\right)\varphi_0 - v\psi_0 + \varepsilon,$$
$$\frac{d\psi_0}{dt} = \left(\frac{a}{2} + w\right)\psi_0 - u\varphi_0 + \delta. \tag{102}$$

As a result of transformation (100)–(102), the last two equations in (5) are reduced to two independent linear equations of the same form:

$$\frac{\partial R}{\partial t} + (az + b)\frac{\partial R}{\partial z} = \nu\frac{\partial^2 R}{\partial z^2},$$
$$\frac{\partial S}{\partial t} + (az + b)\frac{\partial S}{\partial z} = \nu\frac{\partial^2 S}{\partial z^2}. \tag{103}$$

The transformation of the independent variables [51, p. 128]

$$\tau = \int A^2(t)dt + C_1, \quad \zeta = A(t)z + \int A(t)b(t)dt + C_2,$$
$$A(t) = C_3\exp\left(\int a(t)dt\right), \tag{104}$$

where C_1, C_2, and C_3 are arbitrary constants, brings linear equations with variable coefficients (103) to the linear heat conduction equations

$$\frac{\partial R}{\partial \tau} = \nu\frac{\partial^2 R}{\partial \zeta^2}, \quad \frac{\partial S}{\partial \tau} = \nu\frac{\partial^2 S}{\partial \zeta^2}. \tag{105}$$

Remark 1. Formulas (5) and (98)–(105) describe the exact solution of the Navier–Stokes equations that contains ten (twelve) arbitrary functions: six time functions a, b, u, w, ε, and δ and the solutions of two heat conduction equations (105) yield two arbitrary functions if we consider them in the semi-infinite range ζ (if we consider (105) in the finite segment, the solution of each of them yields three arbitrary functions).

Remark 2. To construct a fundamental system of the solutions of homogeneous system (101), it is sufficient to know its one nontrivial particular solution. The solution of inhomogeneous system (102) can be expressed through the fundamental solutions of homogeneous system (101).

CLASSIFICATION OF THE FLOWS

It is convenient to treat the flows described by formulas (4) as the nonlinear superposition of a translational (nonuniform) flow along the z axis and the linear shear flow of a special form. In the vicinity of the point $z = z_0$ that is located at the axis, the components of the velocity of a fluid with allowance for (4) can be expressed as

$$V_k = H_k + G_{km}X_m;$$
$$H_1 = U, \quad H_2 = V, \quad H_3 = F,$$
$$G_{11} = -\frac{1}{2}F_z + w, \quad G_{12} = v, \quad G_{23} = u,$$
$$G_{22} = -\frac{1}{2}F_z - w, \tag{106}$$
$$G_{11} = G_{23} = G_{31} = G_{32} = 0, \quad G_{33} = F_z,$$
$$X_1 = x, \quad X_2 = y, \quad X_3 = z - z_0.$$

Here, $k, m = 1, 2, 3$; G_{km} are the components of the shear flow matrix; summation is taken over the repeating index m; and F_z is the partial derivative with respect to z. All quantities in (106) are taken at $z = z_0$. The equality of the sum of the diagonal elements to zero $G_{11} + G_{22} + G_{33} = 0$ is the corollary of the incompressibility of a fluid.

Any matrix $\|G_{km}\|$ can be represented as the sum of the symmetric and antisymmetric matrixes

Classification of axial flows described by formulas (4) at $U = V = 0$

Flow type	Sought functions	Functions included into pressure
Axisymmetric flow	$u = v = w = 0$	$\alpha = \beta$, $\varepsilon = \delta = \gamma = 0$
Combination of an axisymmetric flow and rotation about the z axis	$w = 0$, $v = -u$	$\alpha = \beta$, $\varepsilon = \delta = \gamma = 0$
Pure deformation flow (without rotation)	$u = v$	α, β, γ – any
General axial flow	$u \neq v$	α, β, γ – any

$$\begin{aligned} \|G_{km}\| &= \|E_{km}\| + \|\Omega_{km}\|, \\ E_{km} &= E_{mk} = \frac{1}{2}(G_{km} + G_{mk}), \\ \Omega_{km} &= -\Omega_{mk} = \frac{1}{2}(G_{km} - G_{mk}). \end{aligned} \tag{107}$$

In turn, by rotating the system of coordinates, the symmetric matrix $\|E_{km}\|$ (in this case, it can be identified with the tensor of deformation rates) can be brought to the diagonal form with the elements E_1, E_2, and E_3, which are the roots of a cubic equation for λ: $\det\|E_{km} - \lambda\delta_{km}\| = 0$, where δ_{km} is the Kronecker symbol.

For the given flow (106), the diagonal elements that determine the intensity of stretching (compressing) motion along the respective axes are calculated by the formulas

$$E_{1,2} = -\frac{1}{2}F_z \pm \frac{1}{2}\sqrt{4w^2 + (u+v)^2}, \quad E_3 = F_z. \tag{108}$$

The partition of the matrix of shear coefficients into the symmetric and antisymmetric parts (107) corresponds to the representation of the field of the velocities of the linear shear flow of a fluid as the superposition of the linear deformation flow with stretch coefficients along the main axes E_1, E_2, and E_3, and the rotation of a fluid as solid at angular velocity $\vec{\omega} = (\Omega_{32}, \Omega_{13}, \Omega_{21})$

For the given flow (106), we have $\Omega_{32} = \Omega_{13} = 0$, and the fluid rotates about the z axis at the angular velocity

$$\Omega_{21} = \frac{1}{2}(u - v). \tag{109}$$

It is easy to show that formulas (108) and (109) remain valid for any point (x_0, y_0, z_0) of considered flow (4).

The analysis of formulas (108) and (109) makes it possible to separate several characteristic types of flows that are specified in the classification table.

ACKNOWLEDGMENTS

This study was supported by the Russian Foundation for Basic Research (project nos. 08-01-00553, 08-08-00530, 07-01-96003-p-ural-a, and 09-01-00343).

NOTATION

F—axial component of the velocity of the fluid;

P—pressure related to the constant density of the fluid;

V_1, V_2, V_3—components of the velocity of the fluid,

t—time;

x, y, z—Cartesian coordinates;

u, v, w, U, V—sought functions in system (5);

a, b, c, m, n, k, A, B, C—constants or functional parameters of the solutions of system (5);

p, q—functional parameters of Eqs. (48) and (75);

f, g, G—sought functions in Eqs. (62), (67), and (75) and their corollaries;

α, β, γ, ε, δ—functional parameters of system (5);

ν—kinematic viscosity of the fluid;

λ, θ, ξ, ζ, ψ, ω—constants or functional parameters of the solutions of Eq. (48) and system (5).

REFERENCES

1. Kutepov, A.M. Polyanin, A.D., et al., *Khimicheskaya gidrodinamika* (Chemical Hydrodynamics), Moscow: Byuro Kvantum, 1996.
2. Polyanin, A.D., Kutepov, A.M., Vyazmin, A.V., and Kazenin, D.A., *Hydrodynamics, Mass and Heat Transfer in Chemical Engineering*, London: Taylor & Francis, 2002.
3. Loitsyanskii, L.G., *Mekhanika zhidkosti i gaza* (Fluid Mechanics), Moscow: Nauka, 1973.
4. Kochin, N.E., Kibel', I.A., and Roze, N.V., *Teoreticheskaya gidromekhanika.* (Theoretical Hydromechanics), Fizmatgiz, 1963.
5. Landau, L.D. and Lifshitz, E.M., *Gidrodinamika* (Hydrodynamics), Moscow: Nauka, 1986.
6. Schlichting, H., *Boundary Layer Theory*, New York: McGraw-Hill, 1974.

7. Pukhnachev, V.V., Symmetries in the Navier–Stokes Equations, *Usp. Mekh.,* 2006, no. 6, pp. 3–76.
8. Drazin, P.G.. and Riley, N., *The Navier-Stokes Equations: A Classification of Flows and Exact Solutions,* Cambridge: Cambridge Univ. Press, 2006.
9. Polyanin, A.D. and Zaitsev, V.F., *Handbook of Nonlinear Partial Differential Equations*, Boca Raton, FL: Chapman & Hall / CRC, 2004.
10. Ludlow, D.K., Clarkson, P.A., and Bassom, A.P., Nonclassical Symmetry Reductions of the Two-Dimensional Incompressible Navier-Stokes Equations, *Stud. Appl. Math.*, 1999, vol. 103, pp. 183–240.
11. Ludlow, D.K., Clarkson, P.A., and Bassom, A.P., Nonclassical Symmetry Reductions of the Three-Dimensional Incompressible Navier-Stokes Equations, *J. Phys., A: Math. Gen.*, 1998, vol. 31, pp. 7965–7980.
12. Ibragimov, N.H., *CRC Handbook of Lie Group to Differential Equations,* Boca Raton, FL: CRC, 1995, vol. 2.
13. Lin, C.C., Note on a Class of Exact Solutions in Magneto-Hydrodynamics, *Arch. Rational Mech. Anal.*, 1958, vol. 1, pp. 391–395.
14. Sidorov, A.F., On Two Classes of the Solutions of the Equations of Fluid Mechanics and Their Relation to the Theory of Traveling Waves, *Prikl. Mekh. Tekh. Fiz.*, 1989, no. 2, pp. 34–40.
15. Meleshko, S.V. and Pukhnachev, V.V., On One Class of the Partially Invariant Solutions of the Navier–Stokes Equations, *Prikl. Mekh. Tekh. Fiz.*, 1999, no. 2, pp. 24–33.
16. Meleshko, S.V., A Particular Class of Partially Invariant Solutions of the Navier-Stokes Equations, *Nonlinear Dyn.*, 2004, vol. 36, no. 1, pp. 47–68.
17. Hagen, G., Uber die Bewegung des Wasser in engen zylindrischen Rohren, *Pogg. Ann.*, 1839, vol. 46, pp. 423–442.
18. Poiseuille, J., Recherches experimentelles sur le mouvement des fluides dans les tubes de tres petits diametres, *Comptes Rendus*, 1841, vol. 12, pp. 112–115.
19. Couette, M., Etudes sur le frottement des fluides, *Ann. Chim. Phys.,* 1890, vol. 21, pp. 433–510.
20. Stokes, G.G., On the Effect of the Internal Friction of Fluid on the Motion of Pendulums, *Trans. Cambridge Philos. Soc.,* 1851, vol. 9, pp. 8–106.
21. Ekman, V.W., On the Influence of the Earth's Rotation on Ocean Currents, *Ark. Mat. Astron. Fys.*, 1905, vol. 2, pp. 1–5.
22. Berker, R., A New Solution of the Navier-Stokes Equations for the Motion of a Fluid Contained between Parallel Plates Rotating about the Same Axis, *Arch. Mech. Stosow*, 1979, vol. 31, no. 2, pp. 265–280.
23. Berker, R., An Exact Solution of the Navier-Stokes Equation: the Vortex with Curvilinear Axis, *Int. J. Eng. Sci*, 1981, vol. 20, no. 2, pp. 217–230.
24. Lai, C.-Y., Rajagopal, K.R., and Szeri, A.Z., Asymmetric Flow between Parallel Rotating Disks, *J. Fluid Mech.*, 1984, vol. 446, pp. 203–225.
25. Polyanin, A.D., Exact Solutions of the Navier–Stokes Equations with the Generalized Separation of Variables, *Dokl. Akad. Nauk*, 2001, vol. 380, no. 4, pp. 491–496 [*Dokl.* Phys. (Engl. Transl.), vol. 46, no. 10, pp. 726–731].
26. Polyanin, A.D. and Zaitsev, V.F., Equations of the Nonstationary Boundary Layer: General Transformations and Exact Solutions, *Teor. Osn. Khim. Technol.*, 2001, vol. 35, no. 6, pp. 563–573.
27. Hiemenz, K., Die Grenzschicht An Einem in Den Gleichformigen Flussigkeitsstrom Eingetauchten Geraden Kreiszylinder, *Dinglers Polytech. J*, 1911, vol. 326, pp. 321–324.
28. Stuart, J.T., The Viscous Flow near a Stagnation Point When the External Flow Has Uniform Vorticity, *J. Aerosp. Sci.*, 1959, vol. 26, pp. 124–125.
29. Tamada, K., Two-Dimensional Stagnation Point Flow Impinging Obliquely on a Plane Wall, *J. Phys. Soc. Jpn.*, 1979, vol. 46, pp. 310–311.
30. Dorrepaal, J.M., An Exact Solution of the Navier-Stokes Equation Which Describes Non-Orthogonal Stagnation Point Flow in Two Dimensions, *J. Fluid Mech.,* 1986, vol. 163, pp. 141–147.
31. Wang, C.Y., Exact Solutions of the Unsteady Navier-Stokes Equations, *Appl. Mech. Rev.*, 1989, vol. 42, no. 11, pp. 269–282.
32. Wang, C.Y., Exact Solutions of the Steady-State Navier-Stokes Equations, *Annu. Rev. Fluid Mech.*, 1991, vol. 23, pp. 159–177.
33. Wang, C.Y., Exact Solutions of the Navier-Stokes Equations - the Generalized Beltrami Flows, Review Extension, *Acta Mech.*, 1990, vol. 81, pp. 69–74.
34. Crane, L.J., Flow past a Stretching Plate, *Z. Angew. Math. Phys.,* 1970, vol. 21, pp. 645–647.
35. Brady, J.F. and Acrivos, A., Steady Flow in a Channel or Tube with an Accelerating Surface Velocity. An Exact Solution to the Navier-Stokes Equations with Reverse Flow, *J. Fluid Mech.*, 1981, vol. 112, pp. 127–150.
36. Berman, A.S., Laminar Flow in Channels with Porous Walls, *J. Appl. Phys.*, 1953, vol. 24, pp. 1232–1235.
37. Terrill, R.M. and Shrestha, G.M., Laminar Flow through Parallel and Uniformly Porous Walls of Different Permeability, *Z. Angew. Math. Phys.*, 1965, vol. 16, pp. 470–482.
38. Novikov, P.A. and Lyubin, L.Ya., *Gidrodinamika shchelevykh sistem* (Hydrodynamics of Slot Systems), Minsk: Nauka i Tekhnika, 1988.
39. von Karman, T., Uber Laminare und Turbulente Reibung, *Z. Angew. Math. Mech.*, 1921, vol. 1, pp. 233–252.
40. Bodewadt, U.T., Die Drehstromung uber Festem Grunde, *Z. Angew. Math. Mech.*, 1940, vol. 20, pp. 241–253.
41. Batchelor, G.K., Note on Class of Solutions of the Navier-Stokes Equations Representing Steady Rotationally - Symmetric Flow, *Q. J. Mech. Appl. Math.,* 1951, vol. 4, pp. 29–41.
42. Stewartson, K., On the Flow between Two Rotating Coaxial Disks, *Proc. Cambridge Philos. Soc.*, 1953, vol. 5, pp. 333–341.
43. Holodniok, M., Kubicek, M., and Hlavacek, V., Computation of the Flow between Two Rotating Coaxial Disk: Multiplicity of Steady-State Solutions, *J. Fluid Mech.*, 1981, vol. 108, pp. 227–240.
44. Brady, J.F. and Durlofsky, L., On Rotating Disk Flow, *J. Fluid Mech.*, 1987, vol. 175, pp. 363–394.

45. Watson, L.T. and Wang, C.Y., Deceleration of a Rotating Disc in a Viscous Fluid, *Phys. Fluids*, 1979, vol. 22, pp. 2267–2269.

46. Lavrent'eva, O.M., Flow of a Viscous Fluid in a Layer on a Rotating Plane, *Prikl. Mekh. Tekh. Fiz.*, 1989, no. 5, pp. 41–48.

47. Agrawal, H.L., A New Exact Solution of the Equations of Viscous Motion with Axial Symmetry, *Q. J. Mech. Appl. Math.*, 1957, vol. 10, pp. 42–44.

48. Terrill, R.M. and Cornish, J.P., Radial Flow of a Viscous, Incompressible Fluid between Two Stationary Uniformly Porous Discs, *Z. Angew. Math. Phys.*, 1973, vol. 24, pp. 676–688.

49. Aristov, S.N., Stationary Cylindrical Vortex in a Viscous Fluid, *Dokl. Akad. Nauk*, 2001, vol. 377, no. 4, pp. 477–480 [*Dokl.* Phys. (Engl. Transl.), vol. 46, no. 4, pp. 251–253].

50. Aristov, S.N. and Gitman, I.M., Viscous Flow between Two Moving Parallel Disks. Exact Solutions and Stability Analysis, *J. Fluid Mech.*, 2002, vol. 464, pp. 209–215.

51. Polyanin, A.D., *Handbook of Linear Partial Differential Equations for Engineers and Scientists*, Boca Raton, FL: Chapman & Hall/CRC, 2002.